MEC2047S · DYNAMICS I · KINEMATICS AND KINETICS (L01–L35)

Dynamics, gently.

A plain-language companion to your lecture notes. It explains what each formula means, why it has the shape it has, and how to draw velocity and acceleration diagrams that earn full marks. Part I covers kinematics (L01–L17); Part II covers kinetics (L19–L35).

Kinematics is the geometry of motion. It asks where things are, how fast they move and how that speed changes, and it never asks *why*. Forces come later, in Part II. Everything in this half of the course comes from one chain of ideas:

$$\text{position} \xrightarrow{d/dt} \text{velocity} \xrightarrow{d/dt} \text{acceleration}$$

Integration walks the chain backwards. The rest of the course is about two choices: **which coordinates** to describe the motion in, and **which reference point** to measure it from. Once you see each topic as one of those two choices, the whole course hangs together.

THE COLOUR CODE IN EVERY DIAGRAM

- **Blue**: an absolute vector you already know completely, such as the velocity of a pin on a driven crank.
- **Green**: a vector perpendicular to a bar. You know its direction but not its size until the diagram closes.
- **Grey dashes**: construction lines. They show a direction before you know where the vector ends.
- **Amber**: a normal component, $\omega^2 r$. Once the velocities are solved, this is always fully known.
- **Crimson**: the absolute vector you are solving for, such as the speed of a slider.

How to use this book

- **Read with your lecture slides open.** Each chapter names the lectures it covers, and most worked examples are the same ones from your slides, slowed right down.
- **Look for the green “Why” boxes.** They explain the science behind a formula, so you can rebuild it in the exam if your memory fails.

- **Use the step-through diagrams.** Click *Next step* (or tap a numbered step) to watch a velocity or acceleration diagram being built one line at a time, in the order you would draw it on paper.
- **Chapters 9 and 10 are the heart of Part I**, and Chapter 20 is where they pay off in Part II. Chapters 11 and 21 are one-page summaries for the night before.

Where each lecture lives

LECTURES	TOPIC	CHAPTER	TUTORIALS
L01	Basics: units, vectors, trig, calculus	1	Tut00
L02–L03	Rectilinear motion and motion graphs	2	Tut01
L04–L08	Curvilinear motion: x–y, n–t, r–θ, 3D	3	Tut02
L09	Relative motion of particles	4	Tut03 Q2
L10	Constrained (pulley) motion	5	Tut03 Q1
L11	Rigid bodies: rotation about a fixed axis	6	Tut03 Q4
L12	Absolute motion analysis	7	Tut03 Q3
L13	Instantaneous centre of zero velocity	8	Tut03 Q5
L14–L15	Relative velocity on a rigid body	9	Tut04
L16–L17	Relative acceleration on a rigid body	10	Tut05
L19–L20	Newton’s second law, straight lines	12	
L21, L24	Newton’s second law, curved paths	13	
L25–L27	Work, energy and power	14	
L28–L29	Linear and angular impulse–momentum	15	
L30	Impact	16	
L32	Mass moment of inertia	17	
L33–L34	Rigid bodies: translation	18	
L32	Rigid bodies: fixed-axis rotation	19	
L33–L35	Rigid bodies: general plane motion	20	

WHY THE DIAGRAMS MATTER SO MUCH

Look at the review lectures. The 2016, 2021 and 2022 exam questions all ask you to *draw* a relative velocity or acceleration vector diagram “to the scale given” and then measure an answer from it. In the 2022 memo, part (b), the acceleration diagram alone, is worth 9 marks. Ticks go to each vector’s magnitude and direction, to the angles, to “all arrows” and to “diagram correct”. This is a drawing skill as much as a maths skill, and it can be learned like a recipe. Chapters 9 and 10 give you that recipe.

CHAPTER 1 · LECTURE L01

The toolkit you will use every day

> In this chapter
>
> - Vectors, and how the course writes them
> - The two triangle rules that solve most vector diagrams
> - The one cross product you need, $\omega\hat{k} \times \vec{r}$
> - Slope, area and the chain rule

L01 is a checklist of maths you are expected to already have. This chapter keeps only the parts that will come up again and again in kinematics, and says *where* each one gets used.

Scalars and vectors

A **scalar** has only a size: time, mass, distance travelled, speed. A **vector** has a size *and* a direction: position, displacement, velocity, acceleration, force. In this course a vector is written with an arrow, $\vec{v}$, and its size (the magnitude) is written without one, v or $|\vec{v}|$.

You will meet two ways of writing the same vector:

$$\vec{v} = \underbrace{(5.32\ \text{m/s},\ \angle -117^\circ)}_{\text{magnitude, angle}} \qquad \text{or} \qquad \vec{v} = \underbrace{-2.44\,\hat{i} + 4.73\,\hat{j}\ \text{m/s}}_{\text{components}}$$

The angle form is the one your memos use for final answers. Angles are measured from the positive x-axis, anticlockwise positive. Converting between them is the bread and butter of the course:

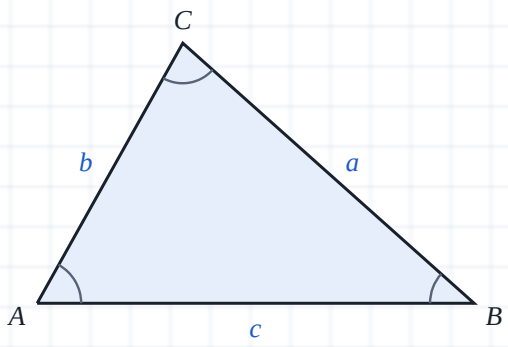
$$v_x = v\cos\theta, \quad v_y = v\sin\theta \qquad \Longleftrightarrow \qquad v = \sqrt{v_x^2 + v_y^2}, \quad \theta = \tan^{-1}\frac{v_y}{v_x}\ \text{(check the}$$

COMMON SLIP

Your calculator's $\tan^{-1}$ only returns angles between −90° and +90°. A vector like $-2.44\hat{i} + 4.73\hat{j}$ points up and to the *left*, so its angle is $180^\circ - 62.7^\circ = 117.3^\circ$, not 62.7°. Sketch the vector before trusting the button.

The two triangle rules

Almost every vector diagram in this course is a triangle. You can measure it with a ruler, or solve it exactly with one of these two rules.



Side a sits opposite angle A, side b opposite B, side c opposite C.

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ **Cosine rule:** $a^2 = b^2 + c^2 - 2bc\cos$

WHICH ONE, WHEN?

Use the **cosine rule** when you know two sides and the angle *between* them. That is the typical relative velocity case: you know v_A, v_B and the angle between their directions, and you want $v_{B/A}$.

Use the **sine rule** when you know all three directions (so all three angles) and one side. That is the typical rigid-body case: one velocity is known completely and the other two are known only by direction. Most of Chapter 9 runs on the sine rule.

The one cross product you need

For flat (planar) motion, every angular velocity points along $\hat{k}$, straight out of the page. Anticlockwise (CCW) is positive. The cross product with a position vector then has a very simple meaning:

$$\omega\hat{k} \times (x\,\hat{i} + y\,\hat{j}) \;=\; \omega\,(-y\,\hat{i} + x\,\hat{j})$$

WHY THIS WORKS

Look at the result: the vector (x, y) has become $(-y, x)$. That is the same vector **turned 90° anticlockwise**, then stretched by ω. So "$\vec{\omega} \times \vec{r}$" just means *turn* $\vec{r}$ *a quarter-turn in the direction of spin, and scale it by* ω. That is why the velocity of a point on a spinning body is always perpendicular to the line from the centre, and why its size is ωr. You will use this idea in Chapters 6, 9 and 10.

The handy unit-vector results are $\hat{k} \times \hat{i} = \hat{j}$ and $\hat{k} \times \hat{j} = -\hat{i}$.

Calculus as pictures

- **A derivative is a slope.** $v = ds/dt$ is the slope of the position–time graph.
- **An integral is an area.** $\Delta s = \int v\, dt$ is the area under the velocity–time graph.
- **The chain rule turns geometry into velocity.** If $x = 0.2 \cos\theta$ and θ changes with time, then $\dot{x} = -0.2 \sin\theta\, \dot{\theta}$. That one line is the whole idea of Chapter 7.

Dots mean time derivatives: $\dot{\theta} = d\theta/dt = \omega$ and $\ddot{\theta} = d^2\theta/dt^2 = \alpha$.

Units and significant figures

- Angles in kinematics formulas are in **radians**. Convert revolutions per minute with $\omega = N \cdot \frac{2\pi}{60}$. For example, 1500 rev/min = 157 rad/s.
- Tutorials ask for **3 significant figures**. Keep 4 or more in your working and round only at the end.
- Check units at the end of every answer. $\omega^2 r$ is (rad/s)²·m = m/s², and ωr is m/s.

IN ONE BREATH

- Vectors carry a size and a direction. Sketch before using $\tan^{-1}$.
- Cosine rule: two sides and the angle between them. Sine rule: one side and all the angles.
- $\omega\hat{k} \times \vec{r}$ means turn $\vec{r}$ 90° the way it spins and multiply by ω.
- Derivative = slope, integral = area, chain rule = geometry into velocity.

CHAPTER 2 · LECTURES L02–L03

Straight-line motion

> In this chapter
>
> - The three defining equations, and where the third one comes from
> - A decision table: what to do when a depends on t, v or s
> - Choosing the limits of integration
> - Reading and drawing motion graphs

A particle moving along a straight line needs only one number to say where it is: its position s, measured from a fixed origin, with one direction chosen as positive. Every idea in the rest of the course is a dressed-up version of this chapter, so it pays to get it rock solid.

Three definitions

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt} \qquad a\,ds = v\,dv$$

The first two are definitions: velocity is how fast position changes, and acceleration is how fast velocity changes. The third is not a new law. It is the second one rewritten with the chain rule:

$$a = \frac{dv}{dt} = \frac{dv}{ds}\cdot\frac{ds}{dt} = \frac{dv}{ds}\,v \quad\Longrightarrow\quad a\,ds = v\,dv$$

WHY WE NEED THE THIRD ONE

Sometimes the acceleration is given as a function of *position*, as with a spring, which pushes harder the further you stretch it. Then $a = dv/dt$ is useless, because you cannot integrate a function of s with respect to t. The chain-rule form swaps time out for position, so you can separate the variables and integrate.

Which equation? Look at what a depends on

The single most useful question in L02–L03 is "*what is the acceleration a function of?*" The answer tells you which equation to separate.

GIVEN	USE	SEPARATE AS	YOU GET
a constant	the constant-a equations	(already integrated)	$v = v_0 + at,\ s = s_0 + v_0 t + \frac{1}{2}at^2,\ v^2 = v_0^2 + 2a(s - s_0)$
$a(t)$	$a = dv/dt$	$\int dv = \int a(t)\,dt$	$v(t)$, then $s(t)$ from $v = ds/dt$
$a(v)$	$a = dv/dt$ *or* $a\,ds = v\,dv$	$\int \frac{dv}{a(v)} = \int dt$ or $\int \frac{v\,dv}{a(v)} = \int ds$	time *or* distance, whichever is asked
$a(s)$	$a\,ds = v\,dv$	$\int a(s)\,ds = \int v\,dv$	$v(s)$

COMMON SLIP

The constant-acceleration equations are only allowed when a really is constant. L03's key outputs say it plainly: *acceleration does not have to be constant*. If a contains t, v or s, you must integrate.

Choosing limits of integration

Each integral has limits on both sides, and the limits on the two sides must describe the **same two moments**. The lower limits are the starting state, and the upper limits are the state you want.

$$\int_{v_0}^{v} dv = \int_{0}^{t} a\,dt \qquad \text{reads as} \qquad \text{“from when the speed was } v_0 \text{ at } t = 0,\ \text{to when it}$$

If you know the end state (say the particle stops, so $v = 0$), put that number in as the upper limit directly.

Worked example 2.1: acceleration that depends on time

`Final exam 2019, Q1 (Review lecture 2)`

A particle is fired into a long test chamber at $v_0 = 27$ m/s when $t = 0$. Its acceleration is $a = -6t$ m/s². Ignoring gravity, how far does it travel before it stops?

1. a depends on t, so start from $a = dv/dt$ and separate:

$$\int_{27}^{v} dv = \int_{0}^{t} -6t\,dt \;\Rightarrow\; v = 27 - 3t^2$$

WHY: Time is the variable we know a in terms of, so time has to be the variable we integrate over.

2. Find when it stops: $0 = 27 - 3t^2 \Rightarrow t = 3$ s.

WHY: "Stops" is a velocity condition, so use the velocity equation to turn it into a time.

3. Integrate again for position:

$$\int_{0}^{x} ds = \int_{0}^{3} (27 - 3t^2)\,dt = \left[27t - t^3\right]_{0}^{3} = 81 - 27$$

WHY: $v = ds/dt$, and v is now a known function of time.

x = 54.0 m

Check: the velocity never changes sign before $t = 3$ s, so distance travelled equals displacement.

Worked example 2.2: acceleration that depends on velocity (drag)

`Same idea as Tut01 Q3–Q4`

A boat cuts its engine at 20 m/s. Water drag gives $a = -0.05\,v^2$ m/s². How far does it coast, and how long does it take, to slow to 10 m/s?

1. **Distance:** the question links speed to distance, so use $a\,ds = v\,dv$:

$$\int_{0}^{s} ds = \int_{20}^{10} \frac{v\,dv}{-0.05v^2} = -20\left[\ln v\right]_{20}^{10} = 20\ln 2$$

WHY: a is a function of v. The $v\,dv$ form gives distance directly, without going through time.

2. **Time:** use $a = dv/dt$:

$$\int_{0}^{t} dt = \int_{20}^{10} \frac{dv}{-0.05v^2} = 20\left[\frac{1}{v}\right]_{20}^{10} = 20(0.1 - 0.05)$$

WHY: Same $a(v)$, but now time is wanted, so put dt on the other side.

s = 13.9 m, t = 1.00 s

Worked example 2.3: constant acceleration `L02 homework, M&K 2/14`

A ball is thrown straight up at 30 m/s from the edge of a 60 m cliff. Find the height h it rises above the edge, and the time to reach the bottom of the cliff.

1. Take up as positive, so $a = -9.81$ m/s², which is constant.

   WHY: Gravity is the only acceleration, and near the Earth it does not change. That is the one situation where the constant-a equations are legal.

2. At the top, $v = 0$: $0 = 30^2 - 2(9.81)h \Rightarrow h = 45.9$ m.

   WHY: We know speeds and want a distance, so use the equation without t in it.

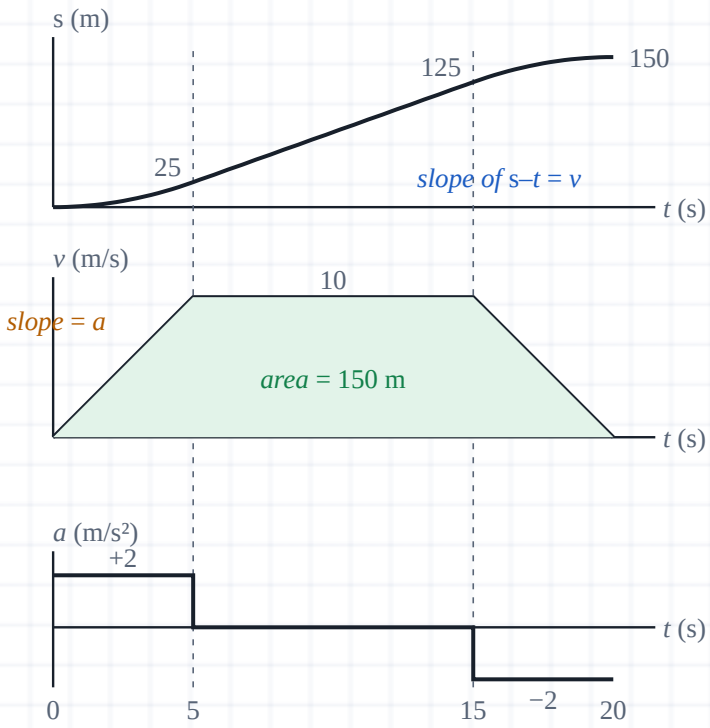
3. The bottom is at $s = -60$ m: $-60 = 30t - 4.905t^2$. Solve the quadratic and keep the positive root: $t = 7.70$ s.

   WHY: Measuring s from the cliff edge means the bottom is *negative*. Signs carry the direction, so there is no need to split the trip into up and down legs.

h = 45.9 m (matches the homework answer), t = 7.70 s

Motion graphs

L03 treats graphs as calculus you can see. Two facts do all the work: the **slope** of a graph is the next quantity down the chain, and the **area** under a graph is the change in the quantity above it.



A trip that speeds up at 2 m/s² for 5 s, cruises at 10 m/s for 10 s, then brakes to rest. Constant a gives a straight v–t line and a parabolic s–t curve. The shaded area under v–t is the distance: 25 + 100 + 25 = 150 m.

FROM THIS GRAPH	SLOPE GIVES	AREA GIVES
s–t	v	
v–t	a	Δs
a–t		Δv
v–s	$a = v\,\frac{dv}{ds}$ (slope times height)	
a–s		$\frac{1}{2}(v^2 - v_0^2)$

WHY THE A–S AREA GIVES ½V²

Integrate $a\,ds = v\,dv$ on both sides: $\int a\,ds = \frac{1}{2}v^2 - \frac{1}{2}v_0^2$. So the area under an a–s graph is not a velocity. It is the change in $\frac{1}{2}v^2$. That is exactly how you build the v–s graph in Tut01 Q2: add up the area as you go, then take a square root.

Drawing a graph from the one above it: a checklist

1. Split the time axis wherever the given graph changes formula.

2. Integrating raises the power by one. A constant becomes a sloped line, and a sloped line becomes a parabola.
3. Carry the end value of each piece into the start of the next. The curve never jumps (unless there is an impact).
4. Label the key values at every break point, and mark maxima. Where $v = 0$, the s–t curve is flat.

IN ONE BREATH

- $v = ds/dt$, $a = dv/dt$, $a\,ds = v\,dv$. The third is the chain rule.
- Ask what a depends on, then separate the matching equation.
- Limits on both sides describe the same two moments.
- Slope goes down the chain, area goes back up it.

CHAPTER 3 · LECTURES L04–L08

Curved paths: three ways to look at the same motion

In this chapter

- Rectangular (x–y) components and projectiles
- Normal–tangential (n–t) components: the key to everything that turns
- Polar (r–θ) components and the strange-looking Coriolis term
- Cylindrical and spherical coordinates for 3D

THE BIG IDEA

The motion is the same whichever coordinates you use. You only choose the coordinate system that makes *your* problem easy. Use **x–y** when the forces line up with fixed directions (gravity). Use **n–t** when you know the path and the speed along it. Use **r–θ** when something is measured from a fixed point, such as a rotating arm, a radar, or a slot.

One fact holds in every system: **velocity is always tangent to the path.** Acceleration usually is not, and the three systems are just three ways of splitting it into pieces.

Rectangular components (L04)

$$\vec{r} = x\hat{i} + y\hat{j} \qquad \vec{v} = \dot{x}\,\hat{i} + \dot{y}\,\hat{j} \qquad \vec{a} = \ddot{x}\,\hat{i} + \ddot{y}\,\hat{j}$$

Because $\hat{i}$ and $\hat{j}$ never change direction, you differentiate each component on its own. A curved motion splits into two straight-line motions, each of which is a Chapter 2 problem.

Projectiles are the classic case. With no air resistance, $a_x = 0$ and $a_y = -g$.

$$x = x_0 + v_{0x}t \qquad v_y = v_{0y} - gt \qquad y = y_0 + v_{0y}t - \tfrac{1}{2}gt^2 \qquad v_y^2 = v_{0y}^2 - 2g(y - y_0)$$

WHY THE HORIZONTAL SPEED NEVER CHANGES

Gravity pulls straight down. Nothing pushes sideways, so nothing changes the sideways velocity. The only thing the horizontal and vertical motions share is the clock. Time is the link you solve for, usually from the vertical equation, and then use in the horizontal one.

Worked example 3.1: a simple projectile

`Tut02 Q1 uses the same method`

A ball is kicked from level ground at 20 m/s, 35° above horizontal. Find its range and maximum height.

1. Split the launch velocity: $v_{0x} = 20\cos 35^\circ = 16.38$ m/s and $v_{0y} = 20\sin 35^\circ = 11.47$ m/s.

2. Time of flight: it lands when $y = 0$ again, so $0 = 11.47t - 4.905t^2 \Rightarrow t = 2.339$ s.

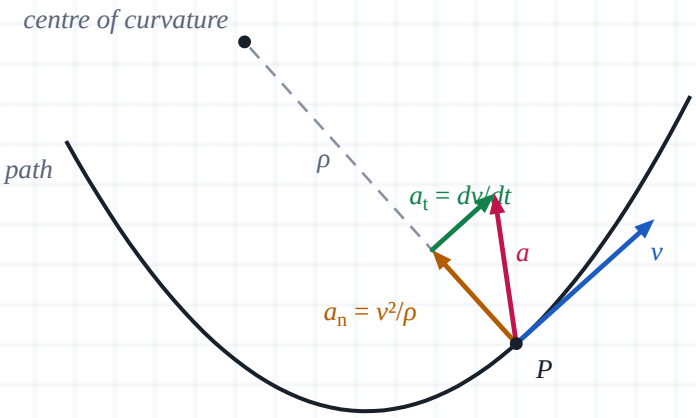
   WHY: The vertical motion decides *when* it lands. The horizontal motion only tells you *where*.

3. Range: $x = 16.38 \times 2.339 = 38.3$ m. Maximum height (where $v_y = 0$): $H = 11.47^2/(2 \times 9.81) = 6.71$ m.

Range 38.3 m, maximum height 6.71 m

Normal and tangential components (L05)

Ride along with the particle. One unit vector $\hat{e}_t$ points forward along the path, and the other, $\hat{e}_n$, points sideways towards the **centre of curvature** (the inside of the bend).



The velocity is along $\hat{e}_t$. The acceleration has a tangential part (speed changing) and a normal part pointing into the bend (direction changing). They add head-to-tail to give $\vec{a}$.

$$\vec{v} = v\,\hat{e}_t \qquad \vec{a} = \underbrace{\dot{v}\,\hat{e}_t}_{a_t} + \underbrace{\frac{v^2}{\rho}\,\hat{e}_n}_{a_n}$$

WHY TWO PIECES, AND WHY V²/P

Velocity can change in two ways: it can get longer or shorter (speed changes), or it can turn (direction changes). Each kind of change needs its own acceleration.

- $a_t = \dot{v} = dv/dt$ changes the **speed**. It is zero when you drive round a bend at a constant speed.
- $a_n = v^2/\rho$ changes the **direction**. It always points into the bend. Going twice as fast round the same bend turns the velocity twice as quickly *and* the velocity is twice as big, so a_n goes up four times. That is why v appears squared. A tighter bend (smaller ρ) turns you faster, so ρ is on the bottom.

Unless the speed is changing, a_n is the only acceleration, which is why you feel pushed outwards in a turning car.

For a path given as $y(x)$, the radius of curvature is $\rho = \frac{[1+(dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ (the hint in Tut02 Q2). On a circle, ρ is just the radius, and that special case drives all of rigid-body rotation (Chapter 6).

Worked example 3.2: a car on a bend `Like L05 Example 1`

A car rounds a bend of radius 50 m at 20 m/s and is speeding up at 3 m/s². Find the size of its acceleration.

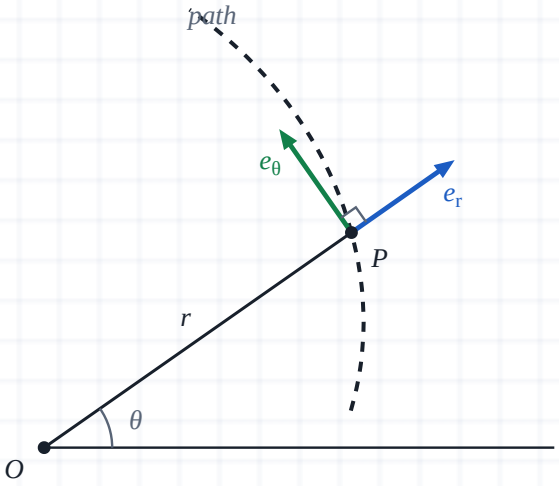
1. $a_t = \dot{v} = 3$ m/s² (given directly as the rate of change of speed).
2. $a_n = v^2/\rho = 20^2/50 = 8$ m/s², pointing to the centre of the bend.
3. The two are perpendicular, so $a = \sqrt{3^2 + 8^2}$.

   WHY: $\hat{e}_t$ and $\hat{e}_n$ are at right angles, so Pythagoras applies.

a = 8.54 m/s²

Polar coordinates (L06–L07)

Now describe the particle by its distance r from a fixed point O and the angle θ of the line OP. The unit vectors are $\hat{e}_r$ (straight out along OP) and $\hat{e}_\theta$ (a quarter-turn anticlockwise from it).



The unit vectors $\hat{e}_r$ and $\hat{e}_\theta$ turn as the particle moves. That turning is what creates the extra terms.

$$\vec{v} = \dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta \qquad \vec{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\hat{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\hat{e}_\theta$$

WHAT EACH TERM MEANS

- $\dot{r}$: sliding outwards along the arm. $r\dot{\theta}$: being carried round by the arm, like a point at radius r on a wheel spinning at $\dot{\theta}$.
- $\ddot{r}$: the outward slide speeding up.
- $-r\dot{\theta}^2$: the centripetal term (the v^2/ρ of a circle). It is negative because it points *inwards*, towards O.
- $r\ddot{\theta}$: the arm spinning faster.
- $2\dot{r}\dot{\theta}$: the **Coriolis** term. Two things happen when you slide outward on a spinning arm. You move to a bigger radius, where the same $\dot{\theta}$ means a faster sideways speed. And your outward velocity is itself being swung round. Each effect contributes one $\dot{r}\dot{\theta}$, so the term is $2\dot{r}\dot{\theta}$.

Worked example 3.3: a collar on a rotating rod

`Same set-up as Tut02 Q3–Q4`

A rod spins at a constant $\dot{\theta} = 2$ rad/s. A collar on it is at $r = 0.5$ m and sliding outward at a steady $\dot{r} = 0.3$ m/s. Find the collar's velocity and acceleration.

1. List the five numbers you need: $r = 0.5$, $\dot{r} = 0.3$, $\ddot{r} = 0$, $\dot{\theta} = 2$, $\ddot{\theta} = 0$.

   WHY: Every polar question comes down to finding these five numbers. The formulas do the rest.

2. $v_r = 0.3$ m/s and $v_\theta = 0.5 \times 2 = 1.0$ m/s, so $v = \sqrt{0.3^2 + 1.0^2} = 1.04$ m/s.

3. $a_r = 0 - 0.5(2)^2 = -2.0$ m/s² (inwards) and $a_\theta = 0 + 2(0.3)(2) = 1.2$ m/s², so $a = 2.33$ m/s².

   WHY: Even with nothing speeding up, there are two accelerations: the inward centripetal one, and the Coriolis one from sliding outward on a spinning rod.

v = 1.04 m/s, a = 2.33 m/s²

3D: cylindrical and spherical (L08)

Cylindrical coordinates are polar coordinates with a height z added. Use them for helices (the roller coaster in L04) and anything turning around a vertical axis while rising.

$$\vec{v} = \dot{r}\,\hat{e}_r + r\dot{\theta}\,\hat{e}_\theta + \dot{z}\,\hat{k} \qquad \vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta + \ddot{z}\,\hat{k}$$

Spherical coordinates (R, θ, ϕ) suit radar antennas and anything pointing into the sky, like latitude (ϕ, measured up from the horizontal plane) and longitude (θ). The velocity, as in L08, is

$$\vec{v} = \dot{R}\,\hat{e}_R + R\dot{\theta}\cos\phi\,\hat{e}_\theta + R\dot{\phi}\,\hat{e}_\phi$$

and the acceleration components are

$$a_R = \ddot{R} - R\dot{\phi}^2 - R\dot{\theta}^2\cos^2\phi \qquad a_\theta = \frac{\cos\phi}{R}\frac{d}{dt}\left(R^2\dot{\theta}\right) - 2R\dot{\theta}\dot{\phi}\sin\phi \qquad a_\phi = \frac{1}{R}\frac{d}{dt}\left($$

You are not expected to derive these. Learn to list $R, \dot{R}, \ddot{R}, \theta, \dot{\theta}, \ddot{\theta}, \phi, \dot{\phi}, \ddot{\phi}$ carefully, just as for polar coordinates, and then substitute.

IN ONE BREATH

- Velocity is always tangent to the path. Choose coordinates to suit the problem.
- x–y: components are independent, and time links them.
- n–t: $a_t = \dot{v}$ changes the speed and $a_n = v^2/\rho$ turns the velocity into the bend.
- r–θ: list the five numbers, then use $(\ddot{r} - r\dot{\theta}^2)$ and $(r\ddot{\theta} + 2\dot{r}\dot{\theta})$.

CHAPTER 4 · LECTURE L09

Relative motion: what one moving observer sees

> In this chapter
>
> - The relative-motion equation and how to read the subscripts
> - Your first vector triangle, drawn step by step
> - Relative acceleration with n–t components

THE BIG IDEA

Picture yourself in car A, watching car B. What you see (B relative to A) is B's real motion with your own motion taken away. Written the other way round, B's real motion is *your* motion plus what you see. That one sentence is the equation for the whole chapter, and for Chapters 9 and 10 as well.

$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} \quad \Longrightarrow \quad \vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \quad \Longrightarrow \quad \vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

Read $\vec{v}_{B/A}$ as "the velocity of B *as seen from* A". The second and third equations are just the first one differentiated, once and twice. In L09 the observer's axes **translate but do not rotate**. The observer slides along with A but keeps facing north.

WHY IT IS JUST ADDITION

Positions add like steps on a map: getting from O to B is the same as getting from O to A and then from A to B. Differentiating a sum gives the sum of the derivatives, so velocities and accelerations add the same way. Nothing more is going on.

Two habits that prevent most mistakes:

- $\vec{v}_{A/B} = -\vec{v}_{B/A}$. Swapping the letters flips the arrow and keeps the size.
- The equation is one vector equation, so it is worth **two** scalar equations (x and y). At most two things can be unknown.

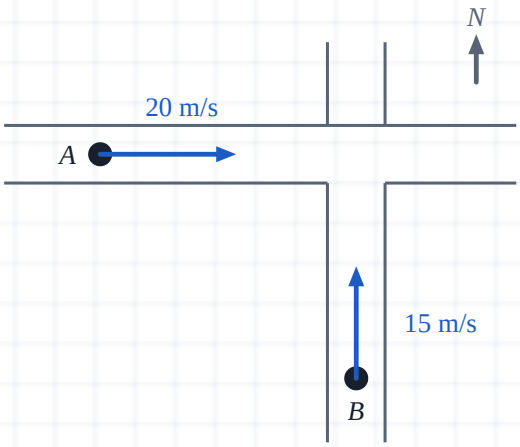
Drawing the vector triangle

The equation $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$ is a triangle. Draw $\vec{v}_A$ first. Put the tail of $\vec{v}_{B/A}$ at the tip of $\vec{v}_A$. The line from the start to the final tip is $\vec{v}_B$. Read it like a journey: go along $\vec{v}_A$, then along $\vec{v}_{B/A}$, and you arrive where $\vec{v}_B$ takes you directly.

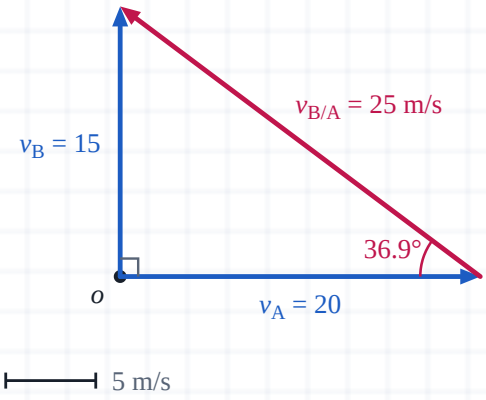
Worked example 4.1: two cars at a crossing

Same method as L09 Examples 1 and 3, Tut03 Q2

Car A drives east at 20 m/s. Car B drives north at 15 m/s on the crossing road. What velocity does B appear to have to the driver of A?



The two cars approach the crossing.



1. Pick a starting point *o* and draw $\vec{v}_A$: 20 m/s east, to scale.
2. From the same point o, draw $\vec{v}_B$: 15 m/s north. Both absolute velocities start at o, because they are both measured from the ground.
3. The vector from the tip of $\vec{v}_A$ to the tip of $\vec{v}_B$ is $\vec{v}_{B/A}$, because $\vec{v}_A + \vec{v}_{B/A}$ must land on the tip of $\vec{v}_B$.

Velocity triangle, drawn to scale.

1. Size, from the right angle at o: $v_{B/A} = \sqrt{20^2 + 15^2} = 25.0$ m/s.

WHY: With no right angle, you would use the cosine rule (Chapter 1).

2. Direction: $\tan^{-1}(15/20) = 36.9°$ north of west. In the course's angle notation that is $\measuredangle 143.1°$.

3. Check with components, which matches:

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = (0, 15) - (20, 0) = (-20, 15)\ \text{m/s}$$

$v_{B/A}$ = 25.0 m/s at 36.9° north of west

It makes sense physically: to the driver of A, the whole world drifts west at 20 m/s, so B appears to move north *and* west.

Relative acceleration

The same triangle works for accelerations. The extra step is that a particle on a **curved** path has two acceleration components (Chapter 3), and you must include both.

$$(\vec{a}_B)_n + (\vec{a}_B)_t = \vec{a}_A + \vec{a}_{B/A}$$

Worked example 4.2: the cars again, with B on a bend

Like L09 Example 3

At the same instant, A is speeding up at 2 m/s². B is on a bend of radius 100 m curving towards the west and is slowing down at 1 m/s². Find $\vec{a}_{B/A}$.

1. $\vec{a}_A = 2$ m/s² east, along its straight path.

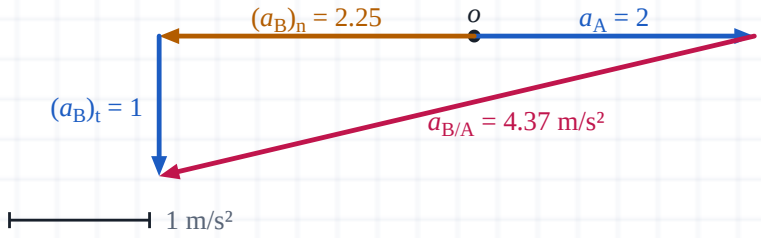
2. $(a_B)_n = v_B^2/\rho = 15^2/100 = 2.25$ m/s², pointing into the bend (west).

   WHY: B is turning, so part of its acceleration must point towards the centre of the bend.

3. $(a_B)_t = 1$ m/s² pointing *south*, opposite to its motion, because B is slowing down.

4. Subtract:

$$\vec{a}_{B/A} = \vec{a}_B - \vec{a}_A = (-2.25, -1) - (2, 0) = (-4.25, -1)\ \text{m/s}^2$$



1. From o, draw the known $\vec{a}_A$: 2 m/s² east.
2. From o, build $\vec{a}_B$ from its parts: $(\vec{a}_B)_n$ west, then $(\vec{a}_B)_t$ south from its tip.
3. Close the triangle from the tip of $\vec{a}_A$ to the tip of $\vec{a}_B$. That closing vector is $\vec{a}_{B/A}$.

Acceleration polygon, drawn to scale.

$a_{B/A}$ = 4.37 m/s² at ∡ −166.8° (13.2° south of west)

IN ONE BREATH

- $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$: B's real motion is A's motion plus what A sees.
- Draw the absolute vectors from one point o. The relative vector joins their tips.
- For a curved path, split acceleration into n (v^2/ρ, into the bend) and t.
- A 2D vector equation can solve for at most two unknowns.

CHAPTER 5 · LECTURE L10

Pulleys: motion tied together by a rope

In this chapter

- The rope-length method (position coordinates)
- The quick “velocity line” method for pulleys
- Why a pulley is secretly a rolling wheel

THE BIG IDEA

A rope does not stretch. Its total length is a constant, so if you write that length in terms of the positions of the blocks and differentiate, the constant disappears. What remains links the velocities, and differentiating again links the accelerations.

Method 1: the rope-length equation

1. **Datums.** Measure each moving thing’s position (s_A, s_B, …) from a *fixed* line, usually through the centre of a fixed pulley. Choose positive in the direction each thing moves (often “down”).
2. **Length.** Add up the rope segments that change length. Leave out bits that never change, such as the arcs wrapped round pulleys. A pulley hanging in a loop of rope contributes its coordinate *twice*, once for each side of the loop.
3. **More than one rope?** Write one length equation per rope, then eliminate the shared coordinate.
4. **Differentiate.** Constants vanish. Watch the signs: a negative answer means opposite to your positive direction.

WHY A PULLEY’S COORDINATE COUNTS TWICE

If a moving pulley hangs in a loop of rope and drops by 1 cm, *both* rope segments beside it get 1 cm longer. That 2 cm has to come from somewhere else in the rope. This is where every factor of 2 in pulley problems comes from: the block on a moving pulley moves at half the speed of the free end.

Worked example 5.1: two ropes `L10 Example 2: keep the slide’s figure open`

The rope at A is pulled down at 2 m/s. Find the speed of block B.

1. Datum through the fixed top pulley, with all coordinates positive downwards. There are three moving things: the end A, pulley C and block B.

2. Rope 1 (A round to C): $s_A + 2s_C = l_1$.

   WHY: Pulley C hangs in a loop of rope 1, so s_C appears twice.

3. Rope 2 (C round B): $s_B + (s_B - s_C) = l_2$.

   WHY: One segment runs from the datum to B. The other runs from C down to B, and its length is the gap $s_B - s_C$.

4. Eliminate s_C: doubling rope 2 gives $4s_B - 2s_C = 2l_2$, and adding rope 1 gives $s_A + 4s_B = l_1 + 2l_2$.

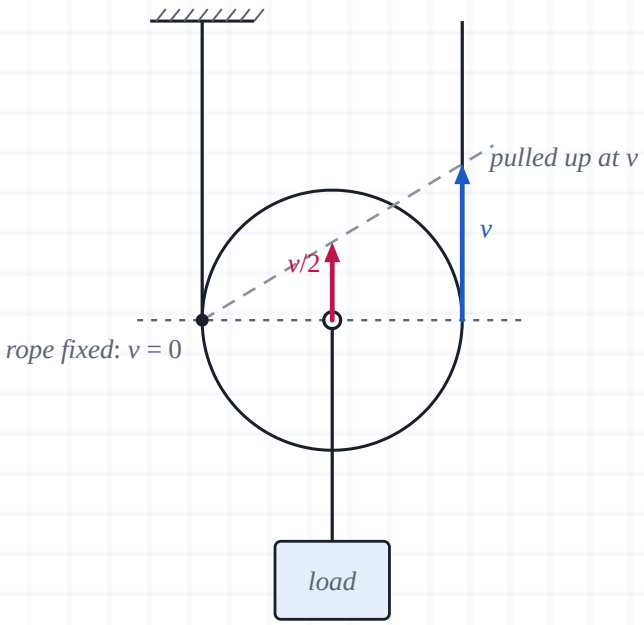
5. Differentiate: $v_A + 4v_B = 0 \Rightarrow v_B = -\frac{1}{4}(2) = -0.5$ m/s.

v_B = 0.5 m/s upwards

The minus sign says B moves against the “positive down” direction, so it rises.

Method 2: the velocity line (velocity analysis)

L10 shows a much faster method that uses one fact: **across a pulley, velocity changes in a straight line**. Draw the pulley with a horizontal diameter, put the rope speeds as arrows at the two edges, and join their tips with a straight line. The height of that line at the centre is the speed of the pulley’s axle.



A movable pulley. The left rope is tied off (speed 0) and the right rope is pulled up at v. The straight line through the two arrow tips crosses the centre at height $v/2$, so the axle and the load rise at half the rope speed.

$$v_{\text{centre}} = \frac{v_{\text{left}} + v_{\text{right}}}{2} \qquad \text{(with signs: up positive)}$$

WHY THE LINE IS STRAIGHT

The rope grips the pulley, so the pulley's rim moves with the rope wherever they touch. A pulley is a rigid disc, so its velocities vary linearly across it, just like a wheel rolling along the rope. If one side of the rope is fixed, that side of the pulley is momentarily still. It is the pulley's **instantaneous centre** (Chapter 8), and the speed grows in proportion to the distance from it.

- **Fixed pulley** (axle fixed): the line passes through the centre, so the two rope sides move at equal speeds in opposite directions.
- **Movable pulley, one side tied:** the centre moves at half the speed of the free side.
- **Both sides moving:** the centre moves at the average. In L10 Example 4, A moves down at 4 m/s and C moves up at 2 m/s, so B moves at $(4 - 2)/2 = 1$ m/s.

Work from the pulley you know most about towards the one you want, carrying each answer across shared ropes.

IN ONE BREATH

- Rope length is constant, so differentiate the length equation.
- A pulley in a loop of rope counts twice, which gives the factors of 2.
- Velocity line: rope speeds at the edges, a straight line between the tips, read off the centre.
- Practice: L10 Examples 3–5 and Tut03 Q1.

CHAPTER 6 · LECTURE L11

Rigid bodies: spinning about a fixed axis

> In this chapter
>
> - The three kinds of planar motion
> - Angular position, velocity and acceleration
> - The three formulas for any point on a spinning body: $v = \omega r$, $a_n = \omega^2 r$, $a_t = \alpha r$
> - Gears and belts

From here on, objects have size. A **rigid body** never changes shape: the distance between any two of its points stays the same. That one property is what makes everything in Chapters 6 to 10 work.

Three kinds of planar motion

- **Translation:** every line on the body stays parallel to itself. It can be *rectilinear* (straight paths) or *curvilinear* (curved paths). Either way, every point has the same velocity and acceleration, so the body behaves like one particle.
- **Rotation about a fixed axis:** every point moves on a circle round the axis (a crank, a gear, a door).
- **General plane motion:** translation and rotation at once (a connecting rod, a rolling wheel, a sliding ladder, a thrown Frisbee). This is where Chapters 7 to 10 live.

Angular motion is Chapter 2 again

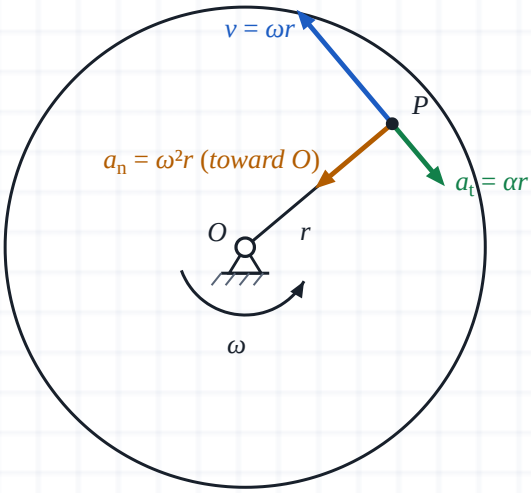
Replace s, v, a by θ, ω, α and every rectilinear result carries straight over:

$$\omega = \frac{d\theta}{dt} \qquad \alpha = \frac{d\omega}{dt} \qquad \omega\, d\omega = \alpha\, d\theta$$

$$\text{constant } \alpha: \quad \omega = \omega_0 + \alpha t \qquad \Delta\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2 \qquad \omega^2 = \omega_0^2 + 2\alpha\,\Delta\theta$$

Anticlockwise is positive. Use the same decision table as in Chapter 2: ask whether α is given as a function of t, ω or θ. (The 2019 exam pulley question gives $\alpha = 0.8\theta$, so it needs $\omega\, d\omega = \alpha\, d\theta$.)

A point on a spinning body



Point P at radius r. Here the body is slowing down (α opposite to ω), so a_t points backwards, against v.

$$v = \omega r \quad (\perp OP) \qquad a_n = \omega^2 r \quad (\text{towards } O) \qquad a_t = \alpha r \quad (\perp OP)$$

WHY THESE THREE

P moves on a circle of radius r, so this is just n–t motion (Chapter 3) with $\rho = r$.

- In one radian of turn, P travels a distance r. Turning at ω rad/s, it travels ωr metres per second.
- $a_n = v^2/\rho = (\omega r)^2/r = \omega^2 r$. It always points to the centre, because the velocity is being turned towards it.
- $a_t = \dot{v} = \frac{d}{dt}(\omega r) = \alpha r$, because r is fixed.

In vector form these are $\vec{v} = \vec{\omega} \times \vec{r}$, $\vec{a}_n = \vec{\omega} \times (\vec{\omega} \times \vec{r}) = -\omega^2 \vec{r}$ and $\vec{a}_t = \vec{\alpha} \times \vec{r}$. These are the Chapter 1 cross product turning $\vec{r}$ a quarter-turn.

COMMON SLIP

If a crank turns at *constant* ω, then $\alpha = 0$ and $a_t = 0$, but $a_n = \omega^2 r$ is **not** zero. A point moving at steady speed round a circle is still accelerating, towards the centre. Forgetting this costs marks in every relative acceleration question.

Gears and belts

Where two wheels touch without slipping, or are joined by a belt, the touching points share the same **speed** and the same **tangential acceleration**:

$$\omega_A r_A = \omega_B r_B \qquad\qquad \alpha_A r_A = \alpha_B r_B$$

Their *normal* accelerations differ, because $\omega^2 r$ depends on each wheel's own ω and r (L17, "bodies in contact").

Worked example 6.1: belt drive starting from rest

`Like L11 Example 1 and Tut03 Q4`

Pulley A (radius 50 mm) starts from rest with $\alpha_A = 6t$ rad/s². A belt drives pulley B (radius 150 mm). At $t = 2$ s, find ω_B, the angle B has turned, and the acceleration of a point on B's rim.

1. α depends on t, so integrate: $\omega_A = \int_0^t 6t\,dt = 3t^2 = 12$ rad/s at $t = 2$, and $\theta_A = \int_0^t 3t^2\,dt = t^3 = 8$ rad.

2. The belt links the rims: $\omega_B = \omega_A r_A / r_B = 12 \times 50/150 = 4.00$ rad/s, and in the same ratio $\theta_B = 8/3 = 2.67$ rad and $\alpha_B = 12/3 = 4$ rad/s².

   WHY: The belt does not stretch or slip, so both rims move the same distance, at the same speed, with the same tangential acceleration.

3. Rim of B: $a_t = \alpha_B r_B = 4(0.15) = 0.60$ m/s² and $a_n = \omega_B^2 r_B = 16(0.15) = 2.40$ m/s², so $a = \sqrt{0.6^2 + 2.4^2}$.

ω_B = 4.00 rad/s, θ_B = 2.67 rad, a = 2.47 m/s²

IN ONE BREATH

- Rigid means fixed distances between points.
- θ, ω, α follow exactly the same rules as s, v, a.
- A point on a spinning body: $v = \omega r$ and $a_t = \alpha r$, both $\perp$ to the radius, plus $a_n = \omega^2 r$ towards the axis.
- Gears and belts share v and a_t at the contact, not a_n.

CHAPTER 7 · LECTURE L12

Absolute motion: write the geometry, then differentiate

> In this chapter
>
> - The four-step tactic from L12
> - The sliding ladder and the slider-crank done by calculus
> - Rolling without slipping: $s = r\theta$

THE BIG IDEA

If you can write *where* a point is as a formula in an angle, you can find how fast it moves by differentiating. The chain rule does the work: every time you differentiate a θ, a $\dot{\theta} = \omega$ appears.

The tactic (from L12)

1. Locate the point with a coordinate s measured from a **fixed** origin.
2. Measure the angle θ of a line in the body from a fixed reference line, and use the body's dimensions to write $s = f(\theta)$.
3. Differentiate once to get v in terms of $\omega = \dot{\theta}$.
4. Differentiate again to get a in terms of ω and $\alpha = \ddot{\theta}$. Use the product rule, because $\dot{\theta}$ can itself change.

Worked example 7.1: the sliding ladder

`The same bar as L14 Example 1, solved by calculus`

A 1.5 m bar has end A on the floor and end B on a wall. A slides away from the wall at 3 m/s. When the bar makes 30° with the floor, find ω and the velocity of B.

1. Geometry, with θ the angle at A between the floor and the bar: $x_A = 1.5\cos\theta$ (from the wall) and $y_B = 1.5\sin\theta$ (up the wall).

   WHY: Both ends are tied to one angle by the fixed length of the bar.

2. Differentiate x_A: $\dot{x}_A = -1.5\sin\theta\,\dot{\theta}$. Put in $\dot{x}_A = 3$: $3 = -1.5(0.5)\dot{\theta} \Rightarrow \dot{\theta} = -4$ rad/s.

   WHY: The minus sign means θ is getting smaller, so the bar is falling flatter, which makes sense as A slides out.

3. Differentiate y_B: $\dot{y}_B = 1.5\cos\theta\,\dot{\theta} = 1.5(0.866)(-4) = -5.20$ m/s.

$|\omega|$ = 4.00 rad/s, v_B = 5.20 m/s down the wall

Keep these numbers in mind. Chapter 8 gets them from a single point, and Chapter 9 from a triangle.

Worked example 7.2: slider-crank `L12 Example 3`

Crank AB (0.2 m) turns at a constant 150 rad/s. Connecting rod BP is 0.75 m, and P slides along the horizontal line through A. Find the velocity of P when $\theta = 30°$.

1. Geometry: $x_P = \underbrace{0.2\cos\theta}_{\text{crank}} + \underbrace{\sqrt{0.75^2 - (0.2\sin\theta)^2}}_{\text{rod, by Pythagoras}}$.

   WHY: The rod's horizontal reach comes from its length and the height of B above the slider line.

2. Differentiate with the chain rule:

$$v_P = -0.2\omega\sin\theta - \frac{0.2^2\sin\theta\cos\theta\,\omega}{\sqrt{0.75^2 - (0.2\sin\theta)^2}}$$

3. Substitute: $v_P = -15.0 - \frac{2.598}{0.7433} = -15.0 - 3.50$.

v_P = 18.5 m/s towards A

Rolling without slipping

A wheel that rolls without slipping lays its rim down on the ground like tape off a roll. After turning through θ, the centre has moved exactly the length of rim that touched the ground:

$$s_O = r\theta \quad\Longrightarrow\quad v_O = r\omega \quad\Longrightarrow\quad a_O = r\alpha$$

L12 Example 4 follows a point on the rim and finds that, at the moment it touches the ground, its velocity is zero but its acceleration is $r\omega^2$ **upwards**. Remember this: it comes back in Chapters 8 and 10.

WHEN TO CHOOSE THIS METHOD

Absolute motion is ideal when the geometry is simple and you want a formula valid at *every* angle (for example a cam follower, L12 Example 2). It becomes painful when the expression is messy, and differentiating it twice for acceleration is where mistakes creep in. For linkages at one instant, the relative velocity diagram (Chapter 9) is usually faster, and it is what the exams ask for.

IN ONE BREATH

- Write position as a function of the angle, using fixed origins.
- Differentiate with the chain rule; every $d\theta/dt$ is ω.
- Rolling without slip: $s = r\theta$, $v = r\omega$, $a = r\alpha$.
- Practice: L12 Questions 1–2 and Tut03 Q3.

CHAPTER 8 · LECTURE L13

The instantaneous centre: pretend it is just spinning

In this chapter

- What the instantaneous centre (IC) is, and why it always exists
- Three ways to find it
- Fast velocity answers from one distance
- The trap: the IC is not a point of zero acceleration

THE BIG IDEA

At any single instant, a body in plane motion moves *as if* it were spinning about one point. That point has zero velocity at that instant, and it is called the instantaneous centre of zero velocity (IC). Find it, and every velocity on the body is just $v = \omega \times (\text{distance to the IC})$, perpendicular to the line to the IC. This is the Chapter 6 formula, borrowed for one moment.

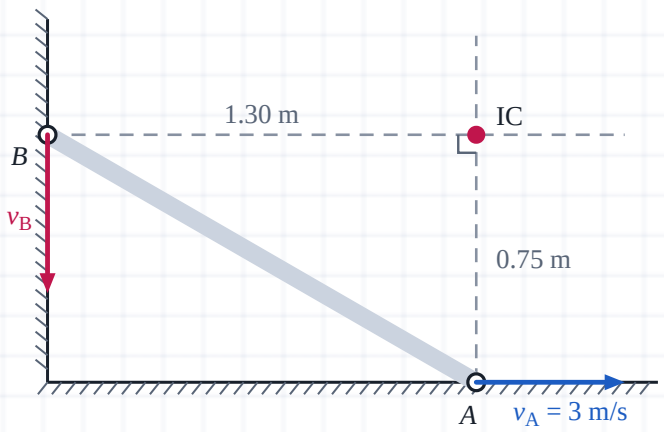
Finding the IC

Every velocity on the body is perpendicular to the line joining that point to the IC. Turned around, the IC must lie on the perpendicular to each velocity. That gives three cases:

1. **You know $\vec{v}_A$ and ω:** the IC is on the perpendicular to $\vec{v}_A$ through A, at distance v_A/ω.
2. **You know the directions of two velocities** (the usual case: a slider, or a pin on a crank): draw the perpendicular to each through its point. The IC is where they cross.
3. **Two parallel velocities, perpendicular to the line joining their points:** use similar triangles. The IC is where the line through the arrow tips meets the line through the points. If the two velocities are equal, the lines never meet: the body is translating, the IC is “at infinity”, and $\omega = 0$.

Worked example 8.1: the sliding ladder in one line

Same bar as Chapters 7 and 9



$\vec{v}_A$ is horizontal, so its perpendicular through A is vertical. $\vec{v}_B$ is vertical, so its perpendicular through B is horizontal. They cross at the IC.

1. Distances from the IC: $r_{A/IC} = 1.5 \sin 30° = 0.75$ m and $r_{B/IC} = 1.5 \cos 30° = 1.30$ m.

2. $\omega = v_A / r_{A/IC} = 3 / 0.75 = 4.00$ rad/s. A moves right while sitting below the IC, so the bar turns anticlockwise about it.

   WHY: Imagine the bar pinned at the IC. For the point below the pin to move right, the bar must turn anticlockwise.

3. $v_B = \omega r_{B/IC} = 4(1.30) = 5.20$ m/s. B is to the left of the IC, so with anticlockwise rotation it moves down.

ω = 4.00 rad/s CCW, v_B = 5.20 m/s ↓ (the same as Chapter 7)

L13 Example 1 does the slider-crank from Chapter 7 the same way and gets $v_P = 18.5$ m/s again. You now have three methods that give one answer, which is a good sign the physics is consistent.

Rolling wheels

A wheel rolling without slip has its IC at the contact point C, because that point is momentarily stuck to the ground. So the centre moves at ωr, and the top of the wheel moves at $\omega(2r) = 2v_O$. The top of a bicycle tyre really does move at twice the bike's speed.

THE TRAP

The IC has zero **velocity**, not zero acceleration. A moment later, a different point is the IC. The contact point of a rolling wheel accelerates at $\omega^2 r$ straight up, towards the centre (L12 Example 4, L17). So **never take moments of acceleration about the IC** and never use $a = \alpha \times (\text{distance to IC})$. For accelerations, use the relative acceleration diagram (Chapter 10).

WHY AN IC ALWAYS EXISTS

Chapter 9 shows that any point's velocity is a reference point's velocity plus $\omega\times$ the relative position. Whenever $\omega \neq 0$, you can always find one position where the spin part exactly cancels the reference velocity. That position is the IC. When $\omega = 0$ there is nothing to cancel with, so there is no IC (it is at infinity) and the body translates.

IN ONE BREATH

- The IC is the point momentarily at rest. Every velocity is $\omega\times$ the distance to it, perpendicular to the line to it.
- Find it where the perpendiculars to two known velocity directions cross.
- Parallel equal velocities put the IC at infinity, so $\omega = 0$.
- Never use the IC for accelerations.
- Practice: L13 Examples 2–4 and Tut03 Q5.

Relative velocity: drawing the velocity diagram

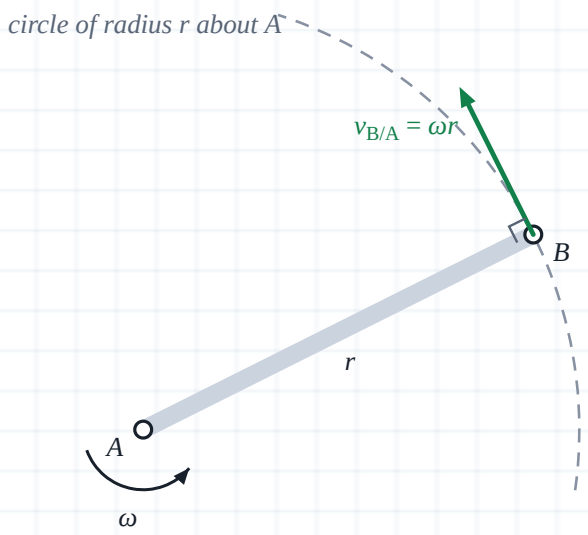
In this chapter

- The one equation, and why the relative velocity is always perpendicular to the bar
- The known/unknown audit that tells you a diagram will close
- A ten-step recipe for drawing the diagram, with the reason for each step
- How to tell clockwise from anticlockwise straight from your diagram
- Four worked examples, three of them past exam questions

The one equation

Take any two points A and B on the same rigid body. Chapter 4 already says $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$. What is new is that, on a rigid body, we know a great deal about $\vec{v}_{B/A}$ before we start:

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \qquad \text{where} \qquad \vec{v}_{B/A} = \vec{\omega} \times \vec{r}_{B/A}, \quad v_{B/A} = \omega r_{AB}, \quad \vec{v}_{B/A} \perp AB$$



An observer riding on A (without turning) sees B swing round on a circle, because the bar's length cannot change. Motion on a circle is always tangent to it, which means perpendicular to the radius AB.

WHY THE RELATIVE VELOCITY IS PERPENDICULAR TO THE BAR

Sit at A, move with it, but do not turn. B is always exactly r_{AB} away from you, because the body is rigid. A point that stays a fixed distance from you can only move around you on a circle. A velocity on a circle is tangent to it, so $\vec{v}_{B/A}$ is perpendicular to AB. Its size is ωr_{AB}, because that is how fast a point on a circle of radius r_{AB} moves when the radius turns at ω (Chapter 6).

So the general plane motion of any rigid body is a **translation** (everything moving with A) plus a **rotation** about A (L14, “interpretation of relative velocity”).

This is why the relative velocity diagram works so well. One of the three vectors always has a **known direction** (perpendicular to the bar) without any calculation. That direction is the green dashed line in every diagram below.

Choose points you know something about

The equation is only useful if A and B are points whose motion you already partly know. In practice these are almost always pins and sliders:

TYPE OF POINT	DIRECTION OF ITS VELOCITY	SIZE
Pin on a crank that turns about a fixed pivot	$\perp$ to the crank	$\omega_{\text{crank}}\, r$ if ω is given, otherwise unknown
Slider (piston, collar, block) in a straight guide	Along the guide	Usually unknown
Contact point of a wheel rolling without slip	Zero, so it makes a great reference point	
Centre of a rolling wheel	Parallel to the surface	ωr
Pin joining two bars	One velocity, shared by both bars. This is how you carry an answer from one bar to the next.	

The audit: will the diagram close?

Every vector has two pieces of information, a **size** and a **direction**. A 2D vector equation gives two scalar equations, so **at most two pieces can be unknown**.

Before drawing anything, fill in a small table. Exam memos award ticks for exactly this table, written next to the equation.

TERM	SIZE	DIRECTION
$\vec{v}_B$ (the reference point, e.g. a pin on a driven crank)	✓ ωr	✓ ⊥ crank
$\vec{v}_{C/B}$	? (it holds ω_{BC})	✓ ⊥ BC
$\vec{v}_C$ (the point you want)	?	✓ along guide or ⊥ its crank

Two question marks, both of them sizes. That is the standard situation, and it means the diagram is a triangle with one fully known side and two known directions.

The recipe: drawing a relative velocity diagram

1. **Draw the mechanism neatly**, roughly to scale, and label every pin. Mark the angles you will need.

   WHY: Every direction in the velocity diagram is copied from this drawing. A sloppy sketch gives sloppy perpendiculars.

2. **Find the bar lengths and angles** from the dimensions (Pythagoras and $\tan^{-1}$).

   WHY: You need each bar's angle to draw its perpendicular, and its length to turn v_{rel} into ω.

3. **Write the equation with the unknown point on the left**: $\vec{v}_{\text{unknown}} = \vec{v}_{\text{known}} + \vec{v}_{\text{unknown/known}}$.

   WHY: Then every term on the right starts from something you know, which fixes the order of drawing.

4. **Fill in the audit table**, size and direction for each term. Calculate any known sizes now, such as $v_B = \omega_{AB} r_{AB}$.

5. **Choose a scale** so that the biggest vector is 8–15 cm long, and write it on the page (for example, "1 m/s = 20 mm").

   WHY: Bigger diagrams give more accurate readings, and the memo awards a mark for using the given scale.

6. **Mark an origin *o*** and draw the fully known vector from it, with an arrowhead and a label.

7. **Through the tip of that vector, draw a long, thin line perpendicular to the bar.** This is the direction of $\vec{v}_{\text{rel}}$. You do not know where it ends yet, so do not put an arrowhead on it.

8. **Through o, draw a long, thin line in the direction the unknown point must move** (along its guide, or perpendicular to its crank).

   WHY: The unknown absolute velocity also starts at o, because every absolute velocity is measured from the same fixed ground.

9. **The two lines cross at the tip of the unknown velocity.** Now draw both arrows: the absolute one from o to the crossing, and the relative one from the tip of the known vector to the crossing. Label everything.

10. **Measure and convert** with your scale (or solve the triangle with the sine rule). Then $\omega_{\text{bar}} = v_{\text{rel}} / r_{\text{bar}}$, and find its sense with the rule below.

Clockwise or anticlockwise? Read it off the diagram

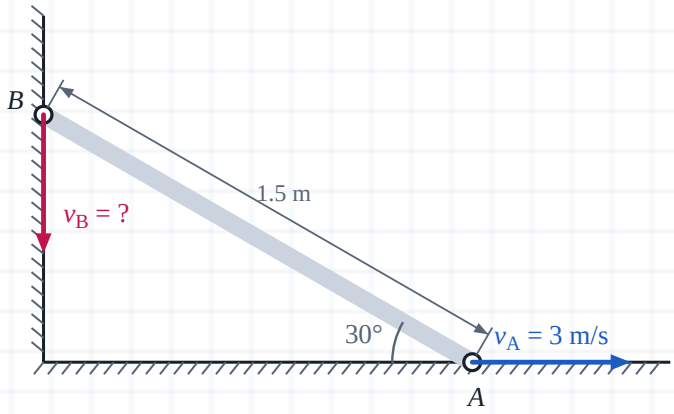
THE PIN-AND-PUSH RULE

$\vec{v}_{C/B}$ is how C moves *as seen from B*. So put your finger on the mechanism at **B** (the letter after the slash) and hold it still. Push the other end, **C**, in the direction of the $\vec{v}_{C/B}$ arrow from your diagram. Whichever way the bar turns, that is the sense of ω_{BC}.

The same rule works for a crank on a fixed pivot: hold the pivot and push the pin in the direction of its velocity. It also works in Chapter 10 for α, using the tangential arrow. If you swap the order ($\vec{v}_{B/C}$ instead of $\vec{v}_{C/B}$), the arrow flips *and* the pinned point swaps, so you get the same answer either way.

Worked example 9.1: the sliding ladder L14 Example 1

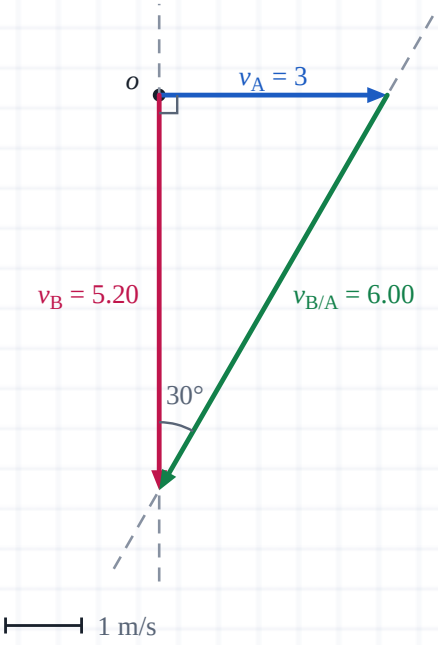
Roller A moves right at 3 m/s. Find the velocity of B and the angular velocity of the 1.5 m bar when $\theta = 30°$.



The mechanism. A is confined to the floor and B to the wall.

Equation: B is the unknown point, so $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$.

TERM	SIZE	DIRECTION
$\vec{v}_A$	✓ 3 m/s	✓ → (floor)
$\vec{v}_{B/A}$	? $= 1.5\,\omega$	✓ ⊥ AB
$\vec{v}_B$	?	✓ vertical (wall)



1. Mark o and draw $\vec{v}_A$ = 3 m/s to the right. It is the only vector known completely, so it goes first.
2. Through the tip of $\vec{v}_A$, draw a line perpendicular to AB. The bar slopes at 30°, so this line is at 30° to the vertical. $\vec{v}_{B/A}$ lies somewhere along it.
3. Through o, draw a vertical line, because B can only move up or down the wall.
4. The lines cross at the tip of $\vec{v}_B$. Draw $\vec{v}_B$ from o, and $\vec{v}_{B/A}$ from the tip of $\vec{v}_A$ to the crossing.

Velocity diagram drawn to scale. Use the buttons, or tap a step, to build it line by line.

1. The triangle has a right angle at o and 30° at the bottom, so $v_B = \frac{v_A}{\tan 30°} = 5.20$ m/s and $v_{B/A} = \frac{v_A}{\sin 30°} = 6.00$ m/s.
2. $\omega = v_{B/A}/r_{AB} = 6.00/1.5 = 4.00$ rad/s.
3. Sense: pin A, and push B along $\vec{v}_{B/A}$, which points down and to the left. B sits up and to the left of A, so pushing it down-left swings the bar anticlockwise.

   WHY: This is the pin-and-push rule. It agrees with the instantaneous centre in Chapter 8.

v_B = 5.20 m/s ↓, ω_{AB} = 4.00 rad/s CCW

CHECK BY VECTOR ALGEBRA (L14, "VECTOR SOLUTION")

Write every vector in components and equate. With $\vec{r}_{B/A} = -1.299\hat{i} + 0.75\hat{j}$:

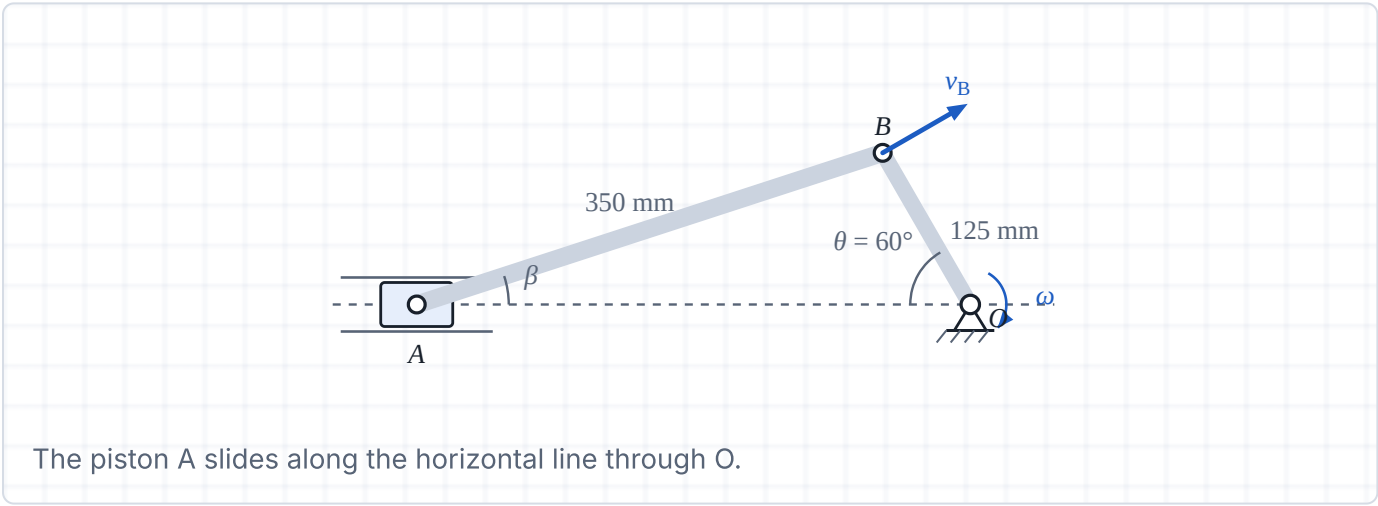
$$-v_B\hat{j} = 3\hat{i} + \omega\hat{k} \times (-1.299\hat{i} + 0.75\hat{j}) = (3 - 0.75\omega)\hat{i} - 1.299\omega\,\hat{j}$$

The $\hat{i}$ parts give $0 = 3 - 0.75\omega$, so $\omega = 4$ (positive means CCW). The $\hat{j}$ parts give $v_B = 1.299(4) = 5.20$. This is the same equation as the diagram, in a different language. It is a good check when you have time.

Worked example 9.2: the slider-crank engine

L15 Example 2 (also the start of L16 Example 2)

Crank OB (125 mm) turns clockwise at 1500 rev/min. Connecting rod AB is 350 mm. Find the piston velocity v_A and ω_{AB} when $\theta = 60°$.



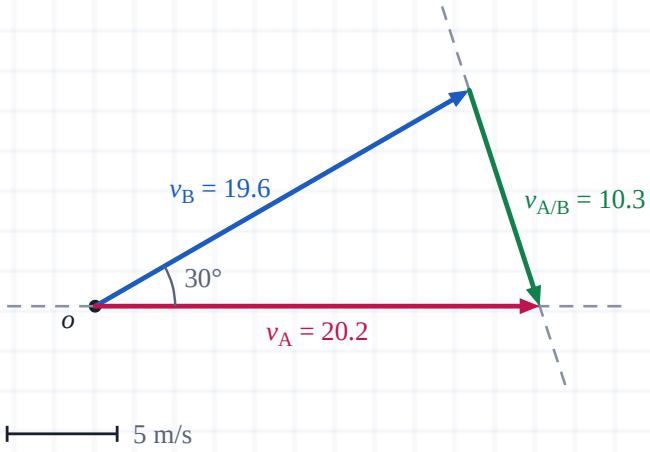
The piston A slides along the horizontal line through O.

1. Known pin: $\omega_{OB} = 1500 \times \frac{2\pi}{60} = 157.1$ rad/s, so $v_B = 157.1 \times 0.125 = 19.6$ m/s, perpendicular to OB. For a clockwise crank with B up and to the left of O, that points up and to the right at 30°.

2. Rod angle: B is $0.125 \sin 60° = 0.108$ m above the slider line, so $\sin\beta = 0.108/0.350$ and $\beta = 18.0°$.

   WHY: You need β to draw the perpendicular to AB in the diagram.

3. Equation, with A unknown: $\vec{v}_A = \vec{v}_B + \vec{v}_{A/B}$. $\vec{v}_B$ is fully known, $\vec{v}_{A/B}$ is ⊥ AB with size unknown, and $\vec{v}_A$ is horizontal with size unknown.



1. From o, draw $\vec{v}_B$ = 19.6 m/s at 30° above horizontal.
2. Through its tip, draw the line perpendicular to AB, which is 18° from the vertical.
3. Through o, draw the horizontal line along which the piston moves.
4. Close the triangle: $\vec{v}_A$ from o to the crossing, and $\vec{v}_{A/B}$ from the tip of $\vec{v}_B$ down to the crossing.

Velocity diagram to scale. The angles inside the triangle are 30° at o, 72° at the piston end and 78° at the tip of $\vec{v}_B$.

4. Sine rule (each side over the sine of the angle opposite it): $\frac{v_A}{\sin 78°} = \frac{v_B}{\sin 72°} = \frac{v_{A/B}}{\sin 30°}$. This gives $v_A = 20.2$ m/s and $v_{A/B} = 10.3$ m/s.

   WHY: All three directions are known, so all three angles are known. That is exactly when the sine rule is the right tool.

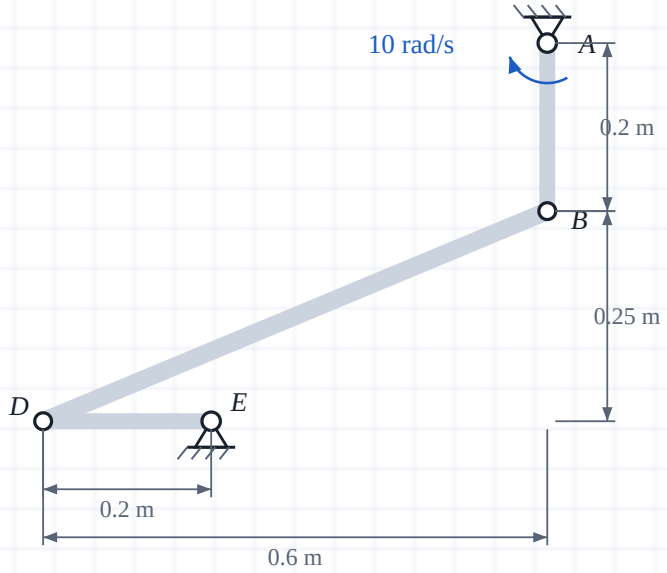
5. $\omega_{AB} = 10.3/0.350 = 29.5$ rad/s. Pin B and push A along $\vec{v}_{A/B}$ (down, slightly right). A is the left end, so the bar turns anticlockwise.

v_A = 20.2 m/s →, ω_{AB} = 29.5 rad/s CCW

Worked example 9.3: three bars, from a past exam

`2016 exam (a)–(b), L17 slides`

Bar AB turns clockwise at a constant 10 rad/s. Draw the relative velocity diagram of bar BD and find the angular velocities of BD and DE.



A and E are fixed pivots. B is shared by AB and BD, and D is shared by BD and DE.

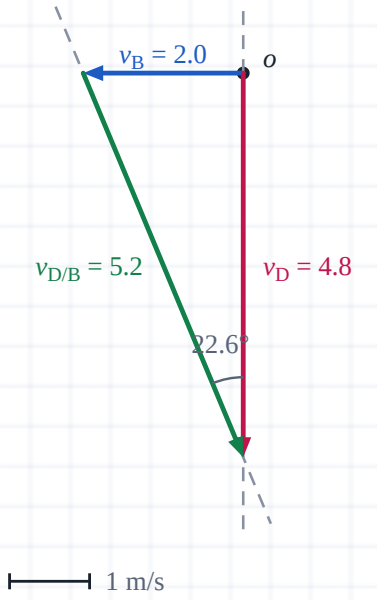
1. Geometry: $r_{BD} = \sqrt{0.25^2 + 0.6^2} = 0.65$ m, and BD rises at $\theta = \tan^{-1}(0.25/0.6) = 22.6°$.

2. Known pin: $v_B = \omega_{AB} r_{AB} = 10(0.2) = 2.00$ m/s. AB is vertical with B below A, so for clockwise rotation B moves **left**.

   WHY: Pin A and turn the bar clockwise: the bottom end swings to the left.

3. D is on DE, which pivots at E. DE is horizontal, so $\vec{v}_D$ is **vertical** (perpendicular to DE).

4. Equation: $\vec{v}_D = \vec{v}_B + \vec{v}_{D/B}$. $\vec{v}_B$ is known completely, $\vec{v}_{D/B}$ is $\perp$ BD (size ?), and $\vec{v}_D$ is vertical (size ?). That is two unknowns, so the triangle will close.



1. From o, draw $\vec{v}_B$ = 2.0 m/s to the left (scale given in the exam: 10 mm = 1 m/s).
2. Through its tip, draw the line perpendicular to BD, at 22.6° from the vertical.
3. Through o, draw a vertical line (the direction of $\vec{v}_D$, ⊥ DE).
4. Close: $\vec{v}_D$ from o down to the crossing, and $\vec{v}_{D/B}$ from the tip of $\vec{v}_B$ to the crossing.

A right-angled triangle, because $\vec{v}_B$ is horizontal and $\vec{v}_D$ is vertical.

5. $v_D = \frac{v_B}{\tan 22.6°} = 4.80$ m/s ↓ and $v_{D/B} = \frac{v_B}{\sin 22.6°} = 5.20$ m/s.
6. $\omega_{BD} = 5.20/0.65 = 8.00$ rad/s. Pin B and push D (down and left of B) towards the lower right: anticlockwise.
7. $\omega_{DE} = v_D/r_{DE} = 4.80/0.2 = 24.0$ rad/s. Pin E and push D (to the left of E) downwards: anticlockwise.

ω_{BD} = 8.00 rad/s CCW, ω_{DE} = 24.0 rad/s CCW (as in the memo)

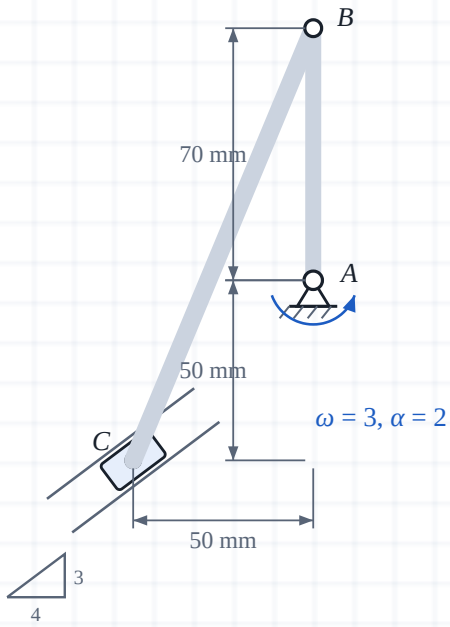
SAME TRIANGLE, DIFFERENT-LOOKING MEMO

The memo writes $\vec{v}_B = \vec{v}_D + \vec{v}_{B/D}$ instead. That is the same equation rearranged: $\vec{v}_{B/D} = -\vec{v}_{D/B}$, so the relative arrow points the other way along the same side of the triangle. Both are marked correct. Pick one form and keep the arrows consistent with it.

Worked example 9.4: a slider on a slope, from a past exam

`Final exam 2022 Q4(a), Review lecture 1`

Bar AB has $\omega_{AB} = 3$ rad/s CCW at this instant. Block C slides on a 3-4-5 incline.
Find the angular velocity of bar BC.

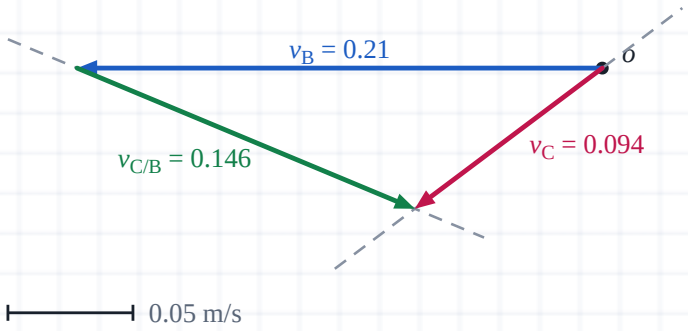


AB is vertical (70 mm). C is 50 mm below and 50 mm to the left of A, so BC = $\sqrt{50^2 + 120^2} = 130$ mm.

1. Known pin: $v_B = 3(0.07) = 0.21$ m/s. B is above A and the rotation is anticlockwise, so B moves **left**.

2. Angles: BC makes $\theta = \tan^{-1}(50/120) = 22.6°$ with the vertical. The slope is at $\beta = \tan^{-1}(3/4) = 36.9°$ to the horizontal.

3. Equation: $\vec{v}_C = \vec{v}_B + \vec{v}_{C/B}$, with $\vec{v}_C$ along the slope (size ?) and $\vec{v}_{C/B} \perp$ BC, which is 22.6° below horizontal (size ?).



1. From o, draw $\vec{v}_B$ = 0.21 m/s to the left.
2. Through its tip, draw the line perpendicular to BC (22.6° to the horizontal).
3. Through o, draw the line along the slope (36.9° to the horizontal).
4. Close: $\vec{v}_C$ from o to the crossing (down the slope), and $\vec{v}_{C/B}$ from the tip of $\vec{v}_B$ to the crossing.

The triangle has angles 36.9° at o, 22.6° at the tip of $\vec{v}_B$, and 120.5° where the lines cross.

4. Sine rule: $\frac{v_{C/B}}{\sin 36.9°} = \frac{v_B}{\sin 120.5°}$, so $v_{C/B} = 0.146$ m/s.
5. $\omega_{BC} = 0.146/0.13 = 1.125$ rad/s. Pin B and push C (below-left of B) towards the lower right: anticlockwise.

ω_{BC} = 1.125 rad/s CCW (memo: 1.125 rad/s CCW)

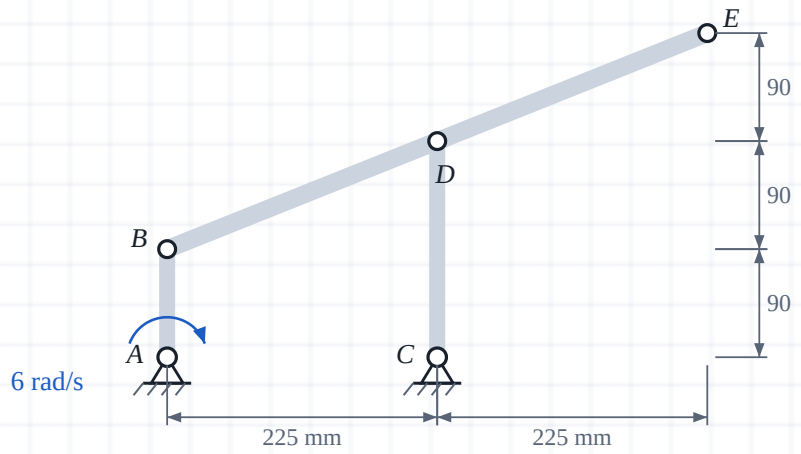
The memo gives a tick for each of: v_B with its direction, the length CB, each of the two angles, the equation, the vector diagram, the arrows, and the final answer with its sense. It is the full recipe above, written down.

A special case: when the diagram collapses

Worked example 9.5: the bar that is not rotating

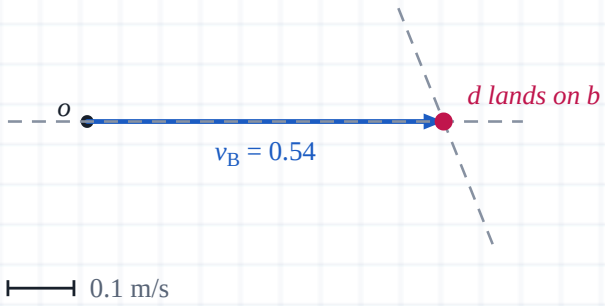
`Final exam 2021 Q5(a), Review lecture 2`

Arm AB turns clockwise at a constant 6 rad/s. Find the angular velocity of bar BDE at the instant shown.



AB (90 mm) and CD (180 mm) are both vertical at this instant.

1. $v_B = 6(0.09) = 0.54$ m/s. B is above A and the rotation is clockwise, so B moves to the right.
2. D is on the vertical arm CD, so $\vec{v}_D$ is horizontal.
3. Equation $\vec{v}_D = \vec{v}_B + \vec{v}_{D/B}$: $\vec{v}_{D/B}$ must be ⊥ BD, which is tilted, while $\vec{v}_D$ and $\vec{v}_B$ are both horizontal.



1. From o, draw $\vec{v}_B$ = 0.54 m/s to the right.
2. Through its tip, draw the line perpendicular to BD.
3. Through o, draw a horizontal line (the direction of $\vec{v}_D$).
4. The lines cross *exactly at the tip of* $\vec{v}_B$, so $\vec{v}_{D/B}$ has zero length.

The "triangle" has collapsed to a single line.

4. $v_{D/B} = 0$, so $\omega_{BDE} = 0$. At this instant BDE is translating, and every point on it moves at 0.54 m/s to the right.

WHY: The perpendiculars to $\vec{v}_B$ and $\vec{v}_D$ are two parallel vertical lines that never meet, so the instantaneous centre is at infinity (Chapter 8).

5 $v_D = 0.54$ m/s, so $\omega_{DC} = 0.54/0.18 = 3.00$ rad/s clockwise.

ω_{BDE} = 0 (instantaneous translation), ω_{DC} = 3.00 rad/s CW

Zero angular velocity does **not** mean zero angular acceleration. In Chapter 10 this same bar has $\alpha = 7.2$ rad/s².

COMMON SLIPS WITH VELOCITY DIAGRAMS

- Drawing $\vec{v}_{\text{rel}}$ *along* the bar instead of perpendicular to it. The bar cannot stretch, so there is never a component along it.
- Starting the unknown absolute velocity from the tip of a vector instead of from o. Absolute velocities all start at o.
- Using the wrong length in $\omega = v_{\text{rel}}/r$. Use the distance between the two points in the equation.
- Forgetting to state the sense (CW or CCW) of each ω. It is a separate mark.
- Not converting rev/min to rad/s.

IN ONE BREATH

- $\vec{v}_C = \vec{v}_B + \vec{v}_{C/B}$, with $\vec{v}_{C/B} \perp$ BC and of size $\omega_{BC}\, r_{BC}$.
- Audit first: at most two unknowns, usually two sizes.
- Known vector from o. Perpendicular line through its tip. Direction line for the unknown point through o. Close at the crossing.
- $\omega = v_{\text{rel}}/r$. Sense by pin-and-push.
- Practice: Tut04 Q1–Q4 (answers are on the sheet), and redo Tut04 with this method as the Relative Velocity page suggests.

Relative acceleration: drawing the acceleration diagram

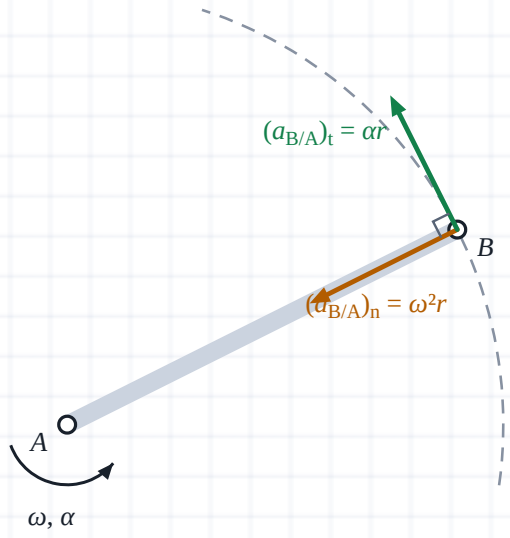
In this chapter

- Why the relative acceleration has two parts, and which part is always known
- The accelerations of pins, sliders and rolling wheels
- The “two chains” recipe for an acceleration diagram that always closes
- Worked examples: ladder, engine, and the 2016, 2021 and 2022 exam questions
- Rolling without slipping, and a slider-crank you can turn by hand

The equation, and why it has two relative parts

Differentiate $\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$ and you get $\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$. Chapter 9 showed that, seen from A, point B moves on a circle of radius r_{AB}. Chapter 3 showed that motion on a circle has two acceleration components. Put those together:

$$\vec{a}_B = \vec{a}_A + (\vec{a}_{B/A})_n + (\vec{a}_{B/A})_t \qquad \begin{cases} (a_{B/A})_n = \omega^2 r_{AB} & \text{from B towards A} \\ (a_{B/A})_t = \alpha\, r_{AB} & \perp AB \end{cases}$$



As seen from A, B runs round a circle. The normal part points along the bar towards A, and the tangential part is perpendicular to the bar.

WHY THE NORMAL PART IS ALWAYS KNOWN (AND WHY VELOCITIES COME FIRST)

The normal part $\omega^2 r$ depends only on ω (from the velocity diagram) and the bar length, and it always points along the bar towards the reference point. So **once the velocity diagram is solved, every normal component in the problem is fully known**: size and direction. That is why the exam always asks for the velocity diagram first, and why the amber arrows in this book are always drawn before the green ones.

The tangential part αr is perpendicular to the bar (direction known) but its size contains the unknown α. It plays exactly the role that $\vec{v}_{B/A}$ played in Chapter 9.

In vector form (L16): $\vec{a}_B = \vec{a}_A - \omega^2 \vec{r}_{B/A} + \vec{\alpha} \times \vec{r}_{B/A}$. The ω^2 term is **not** a cross product. It is ω^2 times $\vec{r}_{B/A}$, with a minus sign, which is just the arrow from B back towards A.

Accelerations of the usual suspects

POINT	NORMAL PART	TANGENTIAL PART
Pin on a crank about a fixed pivot	$\omega^2 r$, towards the pivot (known once ω is known)	αr, $\perp$ crank. **Zero if ω is constant.** Unknown if α is what you want.
Slider in a straight guide	Only one part, along the guide. Size usually unknown. (A straight path has no normal acceleration.)	
Centre of a wheel rolling without slip	$a_O = \alpha r$, parallel to the surface	
Contact point of a rolling wheel	$\omega^2 r$ towards the wheel's centre. Not zero! (Chapter 8's trap.)	
Gear teeth in contact	Different for each gear	Equal: $\alpha_1 r_1 = \alpha_2 r_2$

The recipe: two chains that meet

An acceleration diagram has more arrows than a velocity diagram, but there are still only two unknowns. The trick is to organise the equation so that each unknown comes **last** in a chain of arrows that starts at o.

$$\underbrace{(\vec{a}_C)_n + (\vec{a}_C)_t}_{\text{chain 1: what C does on its own path}} = \underbrace{\vec{a}_B + (\vec{a}_{C/B})_n + (\vec{a}_{C/B})_t}_{\text{chain 2: reach C via B}}$$

1. **Solve the velocity diagram first.** Write down every ω with its sense.

   WHY: Every $\omega^2 r$ term needs these.

2. **Write the equation with the unknown point on the left**, splitting every acceleration on a curved path into n and t. This is the form the memos tick:

   $$(\vec{a}_C)_n + (\vec{a}_C)_t = (\vec{a}_B)_n + (\vec{a}_B)_t + (\vec{a}_{C/B})_n + (\vec{a}_{C/B})_t$$

3. **Audit table:** size and direction of every term. Calculate all the normal parts now. Expect exactly two question marks, both of them sizes of perpendicular terms (or of a slider's acceleration).

4. **Choose and state a scale**, for example 1 mm = 2 m/s².

5. **Chain 2 first:** from o, draw the known parts of $\vec{a}_B$ head to tail, then $(\vec{a}_{C/B})_n$ from its tip, pointing from C towards B.

   WHY: You can only draw a vector with an arrowhead when you know both its size and direction, so all the fully known vectors go down first.

6. **Through the end of chain 2**, draw a long line perpendicular to BC, the direction of $(\vec{a}_{C/B})_t$.

7. **Chain 1:** from o, draw the known part of C's own acceleration. For a pin on a crank that is $(\vec{a}_C)_n$, towards that crank's pivot. For a slider there is no known part, so start at o itself.

8. **Through the end of chain 1**, draw a long line in the unknown direction: ⊥ the crank for $(\vec{a}_C)_t$, or along the guide for a slider.

9. **The two lines cross** at the tip of $\vec{a}_C$. Put arrowheads on both unknown vectors, finishing at the crossing.

   WHY: Both chains describe the same $\vec{a}_C$, so they must end at the same point. The crossing is the only point that lies on both lines.

10. **Measure** the two unknowns, convert with the scale, and get $\alpha = (a_{\text{rel}})_t / r$. Find the sense with pin-and-push on the green arrow. Measure $\vec{a}_C$ itself (o to the crossing) if it is asked for.

NORMAL POINTS TO THE PIN YOU MEASURE FROM

$(\vec{a}_{C/B})_n$ points from C **towards B**, the letter after the slash. For a crank about a fixed pivot P, $(\vec{a}_C)_n$ points from C towards P. Every normal acceleration points inwards, towards the centre the point is circling (relative to that reference point).

Worked example 10.1: the ladder, first acceleration diagram

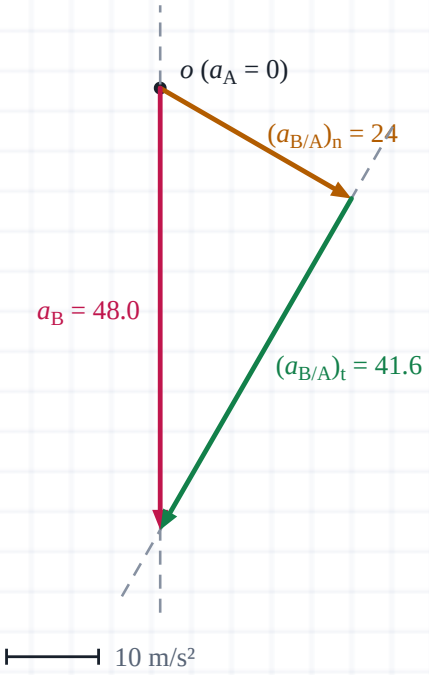
Continues worked example 9.1

The same ladder, with A moving right at a constant 3 m/s (so $\vec{a}_A = 0$). Find $\vec{a}_B$ and α_{AB} at $\theta = 30°$.

1. From Chapter 9: $\omega_{AB} = 4.00$ rad/s CCW.

2. $(a_{B/A})_n = \omega^2 r = 4^2(1.5) = 24.0$ m/s², pointing from B towards A, so down and to the right along the bar.

3. Equation: $\vec{a}_B = \underbrace{\vec{a}_A}_{0} + (\vec{a}_{B/A})_n + (\vec{a}_{B/A})_t$. The unknowns are the size of $\vec{a}_B$ (direction vertical) and the size of $(\vec{a}_{B/A})_t$ (direction $\perp$ AB).



1. Mark o. Because $\vec{a}_A = 0$, chain 2 has nothing to draw for A: it stays at o.
2. Draw $(\vec{a}_{B/A})_n$ = 24 m/s² from o, parallel to the bar, pointing from B towards A.
3. Through its tip, draw the line perpendicular to AB, the direction of $(\vec{a}_{B/A})_t$.
4. Through o, draw a vertical line, because B only moves along the wall.
5. Close: $\vec{a}_B$ from o to the crossing, and $(\vec{a}_{B/A})_t$ from the tip of the amber arrow to the crossing.

Even though A does not accelerate at all, B does, strongly, because the bar is turning.

4. Measured, or from components: $a_B = 48.0$ m/s² and $(a_{B/A})_t = 41.6$ m/s².
5. $\alpha = 41.6/1.5 = 27.7$ rad/s². The green arrow points down-left at B. Pin A and push B that way: anticlockwise.

a_B = 48.0 m/s² ↓, α_{AB} = 27.7 rad/s² CCW

α has the same sense as ω, so the ladder's rotation is speeding up as it falls flatter, which matches what you see when a ladder slips.

Worked example 10.2: the engine's piston acceleration

L16 Example 2 (continues worked example 9.2)

Same slider-crank: crank 125 mm at a constant 1500 rev/min CW, rod 350 mm, $\theta = 60°$. *Find* $\vec{a}_A$ *and* α_{AB}.

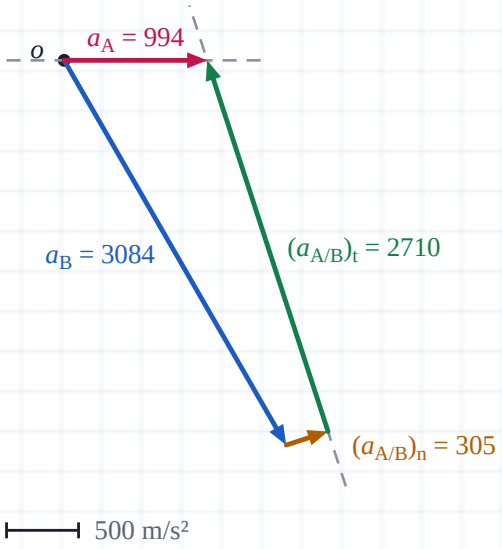
1. From the velocity diagram: $\omega_{OB} = 157.1$ rad/s CW and $\omega_{AB} = 29.5$ rad/s CCW.

2. B is on a crank with *constant* speed, so $(a_B)_t = 0$ and $a_B = (a_B)_n = 157.1^2(0.125) = 3084$ m/s², from B towards O.

3. $(a_{A/B})_n = \omega_{AB}^2 r_{AB} = 29.5^2(0.350) = 305$ m/s², from A towards B (up the rod at 18°).

   WHY: The L16 slide prints 0.250 m here, but 305 only comes out with the real rod length, 0.350 m. That is a typo on the slide; the answer 305 is right.

4. Unknowns: the size of $\vec{a}_A$ (horizontal) and the size of $(\vec{a}_{A/B})_t$ (⊥ AB).



1. From o, draw $\vec{a}_B$ = 3084 m/s² from B towards O (down and to the right at 60°).
2. From its tip, draw $(\vec{a}_{A/B})_n$ = 305 m/s² parallel to the rod, pointing from A to B.
3. Through that tip, draw the line perpendicular to AB.
4. Through o, draw the horizontal line along which the piston moves.
5. Close: $\vec{a}_A$ from o along the horizontal, and $(\vec{a}_{A/B})_t$ up to meet it.

Acceleration diagram to scale (the L16 slide is marked "not to scale").

5. Component check. The y-equation contains only one unknown, because $\vec{a}_A$ is horizontal:

$$y:\ 0 = -3084\sin 60^\circ + 305\sin 18^\circ + 0.951\,(a_{A/B})_t \ \Rightarrow\ (a_{A/B})_t = 2710\ \text{m/}$$

$$x:\ a_A = 1542 + 290 - 0.309(2710) = 994\ \text{m/s}^2$$

WHY: Picking the equation in which one unknown has no component is the fastest way to solve these by hand.

6 $\alpha_{AB} = 2710/0.350 = 7740$ rad/s². The green arrow points up-left at A. Pin B and push A (the left end) upwards: clockwise.

a_A = 994 m/s² →, α_{AB} = 7740 rad/s² CW

Worked example 10.3: the 2016 exam, parts (c) and (d)

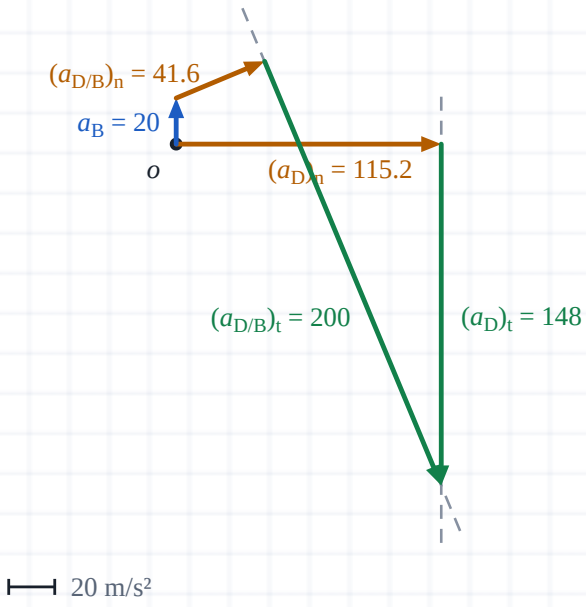
Continues worked example 9.3 · exam scale 1 mm = 2 m/s²

Draw the relative acceleration diagram of bar BD and use it to find α_{BD}. (AB turns at a constant 10 rad/s CW.)

From the velocity diagram: $\omega_{BD} = 8.00$ rad/s CCW and $\omega_{DE} = 24.0$ rad/s CCW. D is the point whose acceleration is not fully known, so put it on the left:

$$(\vec{a}_D)_n + (\vec{a}_D)_t \ = \ \vec{a}_B + (\vec{a}_{D/B})_n + (\vec{a}_{D/B})_t$$

TERM	SIZE	DIRECTION
$(\vec{a}_D)_n = \omega_{DE}^2 r_{DE}$	✓ $24^2(0.2) = 115.2$	✓ D → E (→)
$(\vec{a}_D)_t = \alpha_{DE} r_{DE}$	?	✓ ⊥ DE (vertical)
$\vec{a}_B = \omega_{AB}^2 r_{AB}$ (and $(a_B)_t = 0$)	✓ $10^2(0.2) = 20.0$	✓ B → A (↑)
$(\vec{a}_{D/B})_n = \omega_{BD}^2 r_{BD}$	✓ $8^2(0.65) = 41.6$	✓ D → B (22.6° above →)
$(\vec{a}_{D/B})_t = \alpha_{BD} r_{BD}$	?	✓ ⊥ BD



1. Chain 1: from o, draw $(\vec{a}_D)_n$ = 115.2 m/s² to the right (towards E).
2. Through its tip, draw a vertical line: the direction of $(\vec{a}_D)_t$.
3. Chain 2: from o, draw $\vec{a}_B$ = 20 m/s² upwards (towards A).
4. From its tip, draw $(\vec{a}_{D/B})_n$ = 41.6 m/s², pointing from D towards B (up and to the right, 22.6°).
5. Through that tip, draw the line perpendicular to BD.
6. Close: both green arrows finish at the crossing.

Two chains from o, each ending in a direction line. They meet at the tip of $\vec{a}_D$.

1. Measured: $(a_{D/B})_t \approx 200$ m/s². (Check: the x-components give $115.2 = 0 + 38.4 + 0.385\,(a_{D/B})_t$, so it is 199.7.)
2. $\alpha_{BD} = 200/0.65 = 307$ rad/s². The green arrow points down-right. Pin B and push D (down-left of B) that way: anticlockwise.
3. Bonus: $(a_D)_t = 148$ m/s² ↓, so $\alpha_{DE} = 148/0.2 = 742$ rad/s² CCW.

α_{BD} = 307 rad/s² CCW (memo: 307 rad/s² CCW)

Worked example 10.4: the 2022 exam, parts (b) and (c)

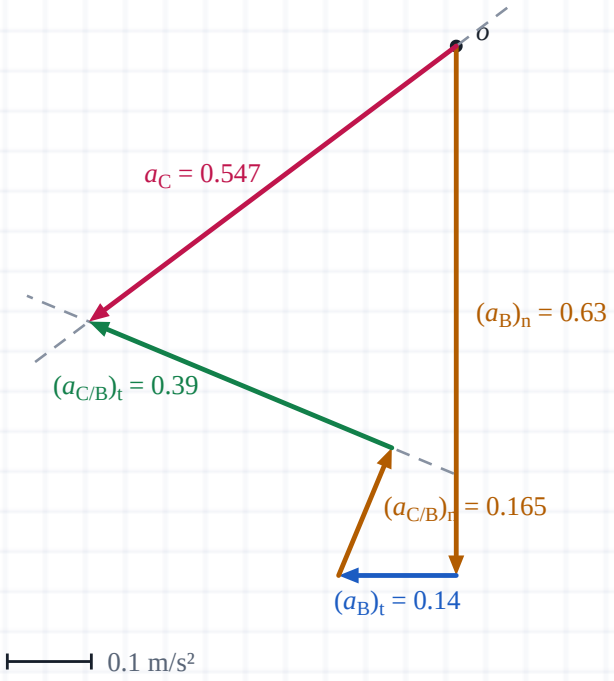
Continues worked example 9.4 · exam scale 1 m/s² : 20 cm

With $\omega_{AB} = 3$ rad/s and $\alpha_{AB} = 2$ rad/s², both CCW, draw the acceleration diagram for bar BC and find the acceleration of block C.

From the velocity diagram: $\omega_{BC} = 1.125$ rad/s CCW. The new feature is that AB is *speeding up*, so B has a tangential part as well.

$$\vec{a}_C = (\vec{a}_B)_n + (\vec{a}_B)_t + (\vec{a}_{C/B})_n + (\vec{a}_{C/B})_t$$

TERM	SIZE (M/S²)	DIRECTION
$(\vec{a}_B)_n = \omega_{AB}^2 r_{AB}$	✓ $9(0.07) = 0.63$	✓ **B → A (↓)**
$(\vec{a}_B)_t = \alpha_{AB} r_{AB}$	✓ $2(0.07) = 0.14$	✓ **⊥ AB, ← (pin A, spin CCW)**
$(\vec{a}_{C/B})_n = \omega_{BC}^2 r_{BC}$	✓ $1.125^2(0.13) = 0.1645$	✓ **C → B (22.6° from vertical)**
$(\vec{a}_{C/B})_t = \alpha_{BC} r_{BC}$	?	✓ **⊥ BC (22.6° from horizontal)**
$\vec{a}_C$	?	✓ **along the slope (36.9°)**



1. From o, draw $(\vec{a}_B)_n$ = 0.63 m/s² downwards (12.6 cm at the exam scale).
2. From its tip, draw $(\vec{a}_B)_t$ = 0.14 m/s² to the left.
3. From that tip, draw $(\vec{a}_{C/B})_n$ = 0.1645 m/s², pointing from C towards B.
4. Through that tip, draw the line perpendicular to BC.
5. Through o, draw the line along the slope, because C is a slider (chain 1 has no known part).
6. Close: $\vec{a}_C$ from o to the crossing, and $(\vec{a}_{C/B})_t$ to the same point.

Chain 2 has three known arrows before its unknown direction line. Chain 1 is just the slope line through o.

1. Measure $\vec{a}_C$ from o to the crossing: 0.547 m/s² (10.9 cm at 20 cm per m/s²). The memo accepts anything between 0.492 and 0.602 m/s², so neat drawing within 5% gets full marks.
2. The green arrow is 0.39 m/s², so $\alpha_{BC} = 0.39/0.13 = 3.0$ rad/s². Pin B and push C up-left: clockwise. (This part was not asked, but it is a useful check.)

a_C = 0.547 m/s², directed down the slope

THE MARK PEOPLE LOSE HERE

Because AB has an angular *acceleration*, $(\vec{a}_B)_t = \alpha_{AB} r_{AB}$ is not zero. The memo gives it its own tick "(mag. & dir.)". Always ask whether the driving crank's ω is constant. If it is not, B has a tangential part.

Worked example 10.5: the 2021 exam, a bar with ω = 0

`Final exam 2021 Q5(b)–(c) · continues worked example 9.5`

AB turns at a constant 6 rad/s CW. Draw the relative acceleration diagram of BDE, find α_{BDE}, then find the acceleration of E.

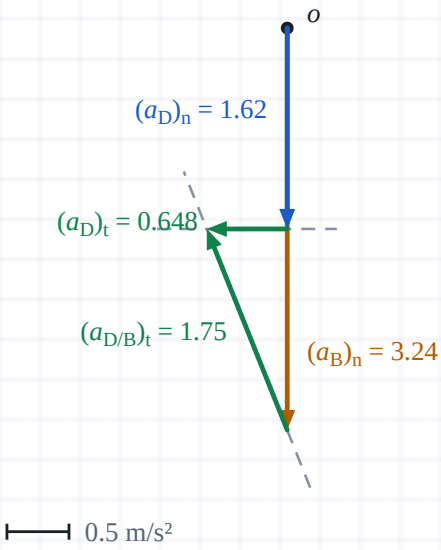
1. From the velocity diagram: $\omega_{BDE} = 0$ and $\omega_{DC} = 3.00$ rad/s CW.

2. Equation for D:

$$(\vec{a}_D)_n + (\vec{a}_D)_t = (\vec{a}_B)_n + (\vec{a}_B)_t + (\vec{a}_{D/B})_n + (\vec{a}_{D/B})_t$$

3. Terms: $(a_D)_n = 3^2(0.18) = 1.62$ ↓ (towards C). $(a_D)_t$: horizontal, size ?.
   $(a_B)_n = 6^2(0.09) = 3.24$ ↓ (towards A). $(a_B)_t = 0$ (constant ω).
   $(a_{D/B})_n = 0^2(0.242) = 0$. $(a_{D/B})_t$: ⊥ BD, size ?.

   WHY: With $\omega_{BDE} = 0$, the normal relative term vanishes. This is the one situation where it does.



1. Chain 2: from o, draw $(\vec{a}_B)_n$ = 3.24 m/s² downwards. The other two parts of chain 2 are zero.
2. Through its tip, draw the line perpendicular to BD.
3. Chain 1: from o, draw $(\vec{a}_D)_n$ = 1.62 m/s² downwards. It lies on top of the first arrow, but is half as long.
4. Through its tip, draw a horizontal line: the direction of $(\vec{a}_D)_t$.
5. Close: both green arrows end at the crossing.

Two downward arrows that overlap are normal in this problem. They belong to different chains.

4. The vertical components give $-1.62 = -3.24 + (a_{D/B})_t \cos 21.8°$, so $(a_{D/B})_t = 1.745$ m/s².

5. $\alpha_{BDE} = 1.745/0.242 = 7.21$ rad/s². The green arrow points up-left at D. Pin B and push D (up-right of B) that way: anticlockwise.

6. Part (c): E is on the same bar, so $\vec{a}_E = \vec{a}_B + (\vec{a}_{E/B})_n + (\vec{a}_{E/B})_t$, with $(a_{E/B})_n = 0$ and $(a_{E/B})_t = 7.21(0.485) = 3.49$ m/s² ⊥ BE. In components:

$$\vec{a}_E = (0, -3.24) + 3.49(-0.371,\ 0.928) = (-1.30,\ 0)\ \text{m/s}^2$$

WHY: Once you know α for a bar, any other point on it follows from one more relative acceleration equation.

α_{BDE} = 7.2 rad/s² CCW, a_E = 1.30 m/s² ← (memo: 7.2 rad/s² and 1.3 m/s² ←)

Rolling without slipping, and gears

For a wheel of radius r rolling on a fixed surface (L17):

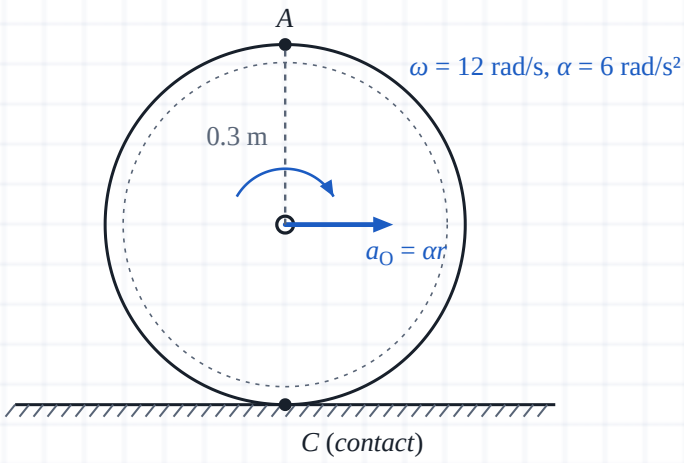
$$v_O = \omega r \qquad a_O = \alpha r \text{ (parallel to the surface)} \qquad a_{\text{contact}} = \omega^2 r \text{ (towards } O\text{)}$$

WHY THE CONTACT POINT ACCELERATES

The contact point is at rest for an instant, but a moment earlier it was coming down and a moment later it is lifting off. Its path is a cusp of a cycloid, and that sharp change of direction needs an acceleration of $\omega^2 r$ straight up into the wheel. This is why you can never use the IC for accelerations.

Worked example 10.6: a gear rolling on a rack `L17 Example 1`

A gear of radius 0.3 m rolls to the right on a fixed rack with $\omega = 12$ rad/s and $\alpha = 6$ rad/s², both clockwise. Find the acceleration of point A at the top.

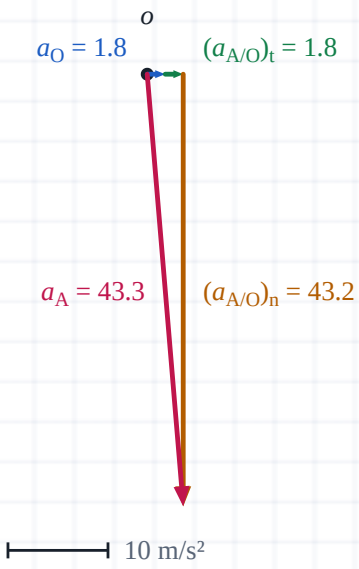


Rolling to the right means clockwise spin.

1. Use the centre O as the reference point, because its acceleration is known:
   $a_O = \alpha r = 6(0.3) = 1.8$ m/s² →.

2. $(a_{A/O})_n = \omega^2 r = 144(0.3) = 43.2$ m/s², from A towards O (↓).
   $(a_{A/O})_t = \alpha r = 1.8$ m/s², ⊥ OA. For CW α at the top of the wheel that is →.

3. Add head to tail: $\vec{a}_A = (1.8 + 1.8)\hat{i} - 43.2\hat{j}$.



1. From o, draw $\vec{a}_O$ = 1.8 m/s² to the right.
2. From its tip, draw $(\vec{a}_{A/O})_t$ = 1.8 m/s², also to the right.
3. From that tip, draw $(\vec{a}_{A/O})_n$ = 43.2 m/s² downwards.
4. $\vec{a}_A$ runs from o to the end of the chain.

Everything is known here, so there is no crossing to find. The normal part dominates.

a_A = 43.3 m/s² at ∠ −85.2°

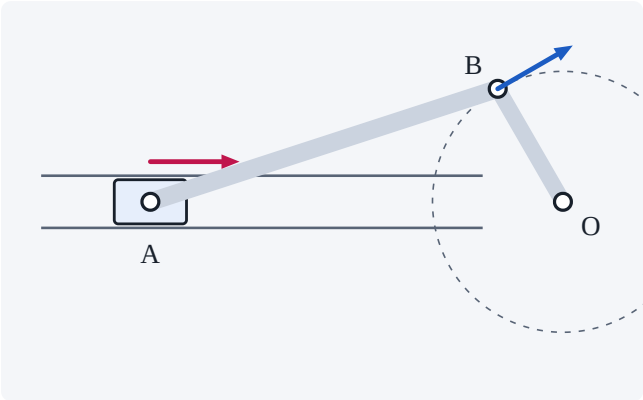
Two gears in mesh share the same a_t at the contact point ($\alpha_1 r_1 = \alpha_2 r_2$), but each has its own $a_n = \omega^2 r$, so their total accelerations at the contact differ (L17, “bodies in contact”).

Lab: turn the crank yourself

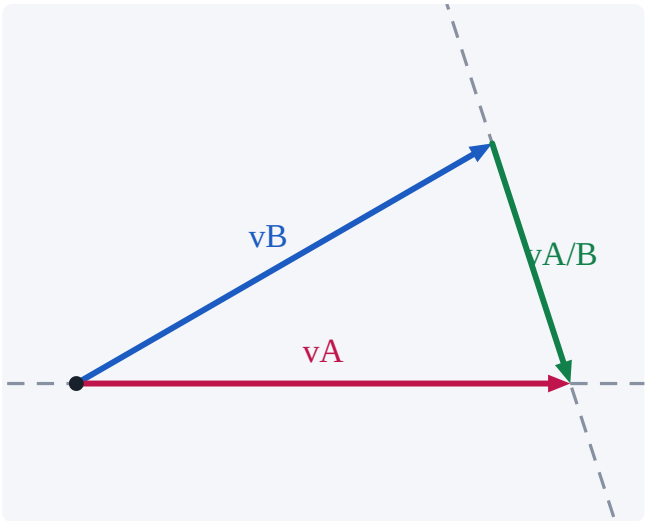
Drag the slider to change the crank angle of the engine from worked examples 9.2 and 10.2. Watch how the two diagrams change shape. Things to look for: at 0° and 180° (dead centres), the piston’s velocity is zero but its acceleration is at its largest. And the green arrow flips direction as the rod’s angular acceleration changes sense.

Crank angle θ **60°**

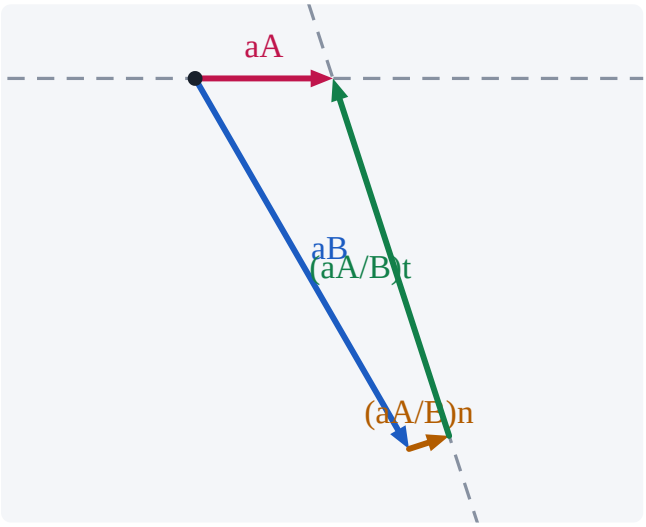
MECHANISM



VELOCITY DIAGRAM



ACCELERATION DIAGRAM



v_A = 20.2 m/s →

ω_{AB} = 29.5 rad/s CCW

a_A = 994 m/s² →

α_{AB} = 7742 rad/s² CW

Each diagram is rescaled to fit its box, so compare shapes, not lengths, between angles. Crank 125 mm, rod 350 mm, 1500 rev/min CW.

COMMON SLIPS WITH ACCELERATION DIAGRAMS

- Starting on accelerations before the velocities are solved. Every normal term needs an ω.
- Pointing a normal arrow the wrong way. It always points *towards* the reference point or the pivot.
- Forgetting $(a_B)_t$ when the driving crank has an α, or putting one in when ω is constant.
- Drawing the tangential term along the bar. It is always $\perp$ to the bar, like the relative velocity.
- Squaring the wrong thing: it is $\omega^2 r$, not $(\omega r)^2$.
- Using the IC to find accelerations.
- Not stating the sense of α (CW/CCW). It is a separate mark.

IN ONE BREATH

- $\vec{a}_C = \vec{a}_B + (\vec{a}_{C/B})_n + (\vec{a}_{C/B})_t$. The normal part is $\omega^2 r$ towards B (known after the velocities). The tangential part is αr, $\perp$ BC (direction only).
- Points on cranks have n and t parts. Sliders have one part, along the guide.
- Two chains from o, unknowns last. Their direction lines cross at the answer.
- $\alpha = (a_{\text{rel}})_t / r$. Sense by pin-and-push on the green arrow.
- Practice: Tut05 Q1–Q5 (answers on the sheet).

Diagram playbook and one-page formula sheet

The diagram question checklist

This list is built from the 2016, 2021 and 2022 memos. Each line matches something the markers tick.

1. **Geometry first.** Bar lengths (Pythagoras) and angles ($\tan^{-1}$). Write them down: they are marks.
2. **Driving crank: $v = \omega r$**, with direction and a sketch arrow. Convert rev/min to rad/s.
3. **Velocity equation** written out, e.g. $\vec{v}_C = \vec{v}_B + \vec{v}_{C/B}$.
4. **Audit**: size and direction of each term, known or not.
5. **Velocity diagram** to the given scale, with arrows on every vector, labels, and angles marked.
6. **ω of each bar** = relative velocity ÷ length, *with sense* (CW or CCW).
7. **Acceleration equation** with every n and t part written separately.
8. **Audit**: every normal part calculated ($\omega^2 r$) with its direction ("towards ___"). Every tangential part as "αr, ⊥ ___, ?".
9. **Acceleration diagram** to scale: known arrows first, two direction lines, crossing, arrowheads on everything.
10. **Measure**, convert, then α = tangential ÷ length, *with sense*. Answers within about ±5% earn full marks.

Choosing a scale when none is given

Find the largest known vector and aim for 10–15 cm. For example, 0.63 m/s² at 1 m/s² : 20 cm is 12.6 cm. 3084 m/s² at 1 mm = 25 m/s² is 123 mm. Always write the scale beside the diagram. Draw thin, sharp lines; use set squares for perpendiculars and a protractor for angles. Practise with the course app (convect.uct.ac.za) and then on paper, because the exam is on paper.

If you get stuck

- **Count the unknowns.** More than two means you have missed a known direction (a slider's guide, a crank's perpendicular) or need another equation for a different bar.
- **Check the IC** for a fast sanity check of any ω (Chapter 8).
- **Switch to components** (vector algebra) if the diagram looks wrong. Pick the component direction in which one unknown disappears.
- **Parallel velocities** on a bar mean $\omega = 0$ for that bar, so its normal relative term is zero, but not its α.

Formula sheet: Part I on one page

TOPIC	FORMULAS	REMEMBER
Rectilinear	$v = \dot{s},\ a = \dot{v},\ a\,ds = v\,dv$ Constant a: $v = v_0 + at,\ s = s_0 + v_0 t + \frac{1}{2}at^2,\ v^2 = v_0^2 + 2a\Delta s$	Ask what a depends on. Constant-a equations only if a is constant.
Graphs	Slope of s–t = v, slope of v–t = a, area under v–t = Δs, area under a–s = $\frac{1}{2}\Delta(v^2)$	Integrating raises the curve's power by one.
Projectile	$a_x = 0,\ a_y = -g$	Time links x and y.
n–t	$a_t = \dot{v},\ a_n = v^2/\rho$	a_n points into the bend.
Polar	$v_r = \dot{r},\ v_\theta = r\dot{\theta}$ $a_r = \ddot{r} - r\dot{\theta}^2,\ a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$	List $r, \dot{r}, \ddot{r}, \dot{\theta}, \ddot{\theta}$ first.
Relative (particles)	$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A},\ \vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$	Non-rotating observer. $\vec{v}_{A/B} = -\vec{v}_{B/A}$.
Pulleys	$\sum(\text{segment lengths}) = \text{const}$ → differentiate. Centre speed = average of the side speeds.	A pulley in a loop counts twice.
Fixed-axis rotation	$\omega = \dot{\theta},\ \alpha = \dot{\omega},\ \omega\,d\omega = \alpha\,d\theta$ $v = \omega r,\ a_n = \omega^2 r,\ a_t = \alpha r$	ω [rad/s] = rev/min × 2π/60. Contacts share v and a_t.
Rolling (no slip)	$v_O = \omega r,\ a_O = \alpha r,\ a_{\text{contact}} = \omega^2 r$ (towards O)	Contact point: $v = 0$, $a \neq 0$.
IC	$v = \omega \cdot r_{\text{to IC}}$, ⊥ to the line to the IC	Velocities only. Never accelerations.
Relative velocity (rigid)	$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{r}_{B/A},\ v_{B/A} = \omega r_{AB} \perp$ AB	Known vector from o; ⊥ line through its tip; direction line through o.
Relative acceleration (rigid)	$\vec{a}_B = \vec{a}_A - \omega^2 \vec{r}_{B/A} + \vec{\alpha} \times \vec{r}_{B/A}$ $(a_{B/A})_n = \omega^2 r$ (B → A), $(a_{B/A})_t = \alpha r$ (⊥ AB)	Velocities first. Two chains from o, unknowns last.

THE WHOLE OF PART I IN FOUR SENTENCES

Velocity is the slope of position, and acceleration is the slope of velocity. On a curve, acceleration splits into a part that changes speed and a part, v^2/ρ, that turns you into the bend. On a rigid body, any point's motion is a reference point's motion plus a spin about it, and that spin always moves the point perpendicular to the line joining them. Draw that as a triangle for velocities and a polygon for accelerations, and you can solve every mechanism in this course.

PART II · KINETICS · LECTURES L19–L35

Part II: why things move

Part I described motion without asking what causes it. Part II brings in forces and masses. The good news is that every acceleration, every v^2/ρ, every $r\ddot{\theta}$ and every relative acceleration diagram from Part I is reused here. Kinetics is kinematics plus one law.

$$\sum \vec{F} = m\vec{a}$$

Everything else in Part II is this law in a different outfit. Integrate it along the path and you get the **work–energy** principle. Integrate it over time and you get **impulse–momentum**. Apply it to a body with size and you add a moment equation, $\sum M_G = I_G\alpha$.

THE COLOUR CODE IN PART II DIAGRAMS

- **Blue**: a force you know, such as the weight or a given push.
- **Amber**: an $m\vec{a}$ term on the kinetic diagram. It is the *effect* of the forces, not another force.
- **Crimson**: an unknown force, such as a reaction, tension, normal force or friction.
- **Green curl**: an $I_G\alpha$ term on the kinetic diagram (rigid bodies only).

Which tool? The first decision in every kinetics question

THE QUESTION ASKS FOR...	USE	BECAUSE
A force, tension, reaction or acceleration *at one instant*	**Newton's second law** $\sum F = ma$ (and $\sum M_G = I_G\alpha$)	It links forces to acceleration at that moment.
A speed at one *position*, given the speed at another position	**Work–energy** $T_1 + V_1 + U'_{1-2} = T_2 + V_2$	It is a scalar equation with no time in it. Forces that do no work drop out.
A speed after a *time*, or an average force during a short hit	**Impulse–momentum** $m\vec{v}_1 + \sum \int \vec{F}\,dt = m\vec{v}_2$	It deals with time and forces that act briefly.
Speeds just after a collision	**Conservation of momentum** plus the coefficient of restitution e	The impact forces are internal and huge, but they cancel in pairs.
A spin rate after something moves in or out, or an impact on a pivoted body	**Angular momentum** about a fixed point	The pivot force has no moment about the pivot.

Many exam questions chain two tools together. A typical chain is energy (falling), then momentum (impact), then energy again (compressing a spring), as in the putty question in Chapter 16. Spot where one stage ends and the next begins.

CHAPTER 12 · LECTURES L19–L20

Newton’s second law in a straight line

> In this chapter
>
> - The free-body diagram and the kinetic diagram
> - The four-step strategy from L19
> - Friction: first ask “will it move?”
> - Connected bodies, pulleys and forces that depend on speed

Newton’s three laws, in one line each

1. **First law:** with no net force, a particle stays at rest or keeps moving in a straight line at constant speed.
2. **Second law:** a net force causes an acceleration in the same direction, $\sum \vec{F} = m\vec{a}$.
3. **Third law:** two bodies push on each other with equal and opposite forces along the same line.

The first and third laws were the whole of statics. The second law is the whole of dynamics. It only fails near the speed of light or at atomic sizes, which never come up here.

The free-body diagram and the kinetic diagram

Draw the body on its own, cut free from everything it touches. Then put on it **every force** that acts on it:

- its **weight** $W = mg$, straight down from the mass centre;
- a **normal force** N at each contact, perpendicular to the surface;
- **friction** F_f at each rough contact, along the surface;
- **tensions** along each rope, pulling away from the body;
- any **applied** force.

Next to it, draw the **kinetic diagram**: the same body with only one arrow, $m\vec{a}$, in the direction the body accelerates. Newton’s second law says the two pictures are equivalent: the forces on the left add up to the $m\vec{a}$ on the right.

WHY TWO PICTURES (AND WHY YOUR NOTES USE ONE)

L19 draws the "inertial force" $m\vec{a}$ on the same diagram as the forces. That is fine, but it tempts you to treat $m\vec{a}$ as one more force and add it to the others. Drawing it on its own picture, with an equals sign between, makes the equation $\sum F = ma$ visible: sum everything on the left, and set it equal to what is on the right, component by component.

The strategy (L19)

1. **Choose axes** that line up with the acceleration (along and across a slope, for example).

   WHY: Then one equation contains the acceleration and the other has $ma = 0$.

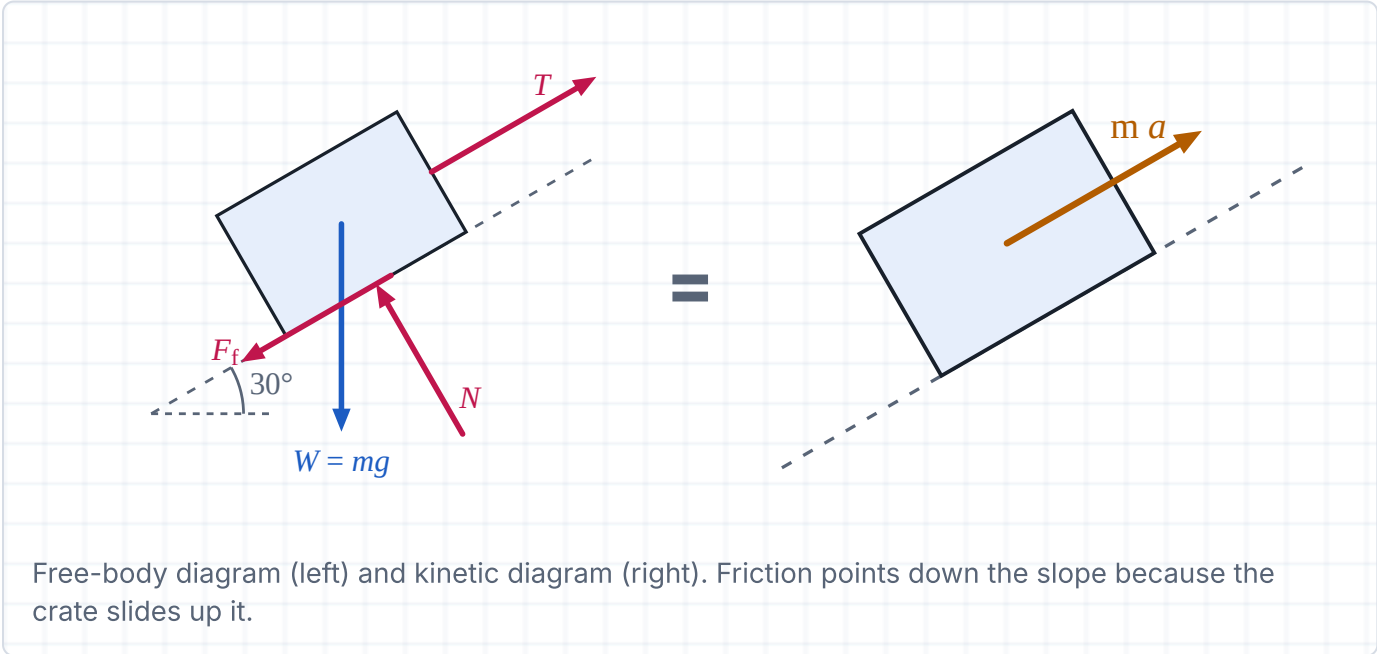
2. **Draw the FBD and the kinetic diagram.**

3. **Write** $\sum F_x = ma_x$ **and** $\sum F_y = ma_y$, stating which way is positive for each.

4. **Bring in kinematics** if needed: a constant-a equation, a rope constraint (Chapter 5), or $a\,ds = v\,dv$.

Worked example 12.1: dragging a crate up a slope `L19 Example 2`

A motor winds in a cable with constant acceleration so that a 20 kg crate moves 6 m up a 30° incline in 3 s, starting from rest. $\mu_k = 0.3$. *Find the cable tension.*



Free-body diagram (left) and kinetic diagram (right). Friction points down the slope because the crate slides up it.

1. Kinematics first: $s = \frac{1}{2}at^2 \Rightarrow 6 = \frac{1}{2}a(3)^2$, so $a = 1.33$ m/s² up the slope.

WHY: The question gives a distance and a time, which is a constant-acceleration statement. Find a before touching the forces.

2. Across the slope ($a_y = 0$): $N - mg\cos 30° = 0$, so $N = 169.9$ N.
3. Friction while sliding: $F_f = \mu_k N = 50.97$ N.
4. Along the slope, up positive: $T - F_f - mg\sin 30° = ma$, so $T = 50.97 + 98.1 + 26.67$.

   WHY: Three things must be paid for: friction, the slope component of the weight, and the acceleration itself.

T = 176 N

Friction: first ask "will it move?"

Static friction is a *reaction*. It is only as big as it needs to be, up to a limit of $\mu_s N$. Kinetic friction, once sliding, is fixed at $\mu_k N$. So L20 starts every friction question the same way:

1. Assume the body stays still ($a = 0$) and work out the friction force that would be **needed**.
2. Compare it with the **maximum available**, $\mu_s N$.
3. If needed ≤ available, nothing moves and $a = 0$. If needed > available, it slides: use $F_f = \mu_k N$, opposite to the sliding.

COMMON SLIP

Writing $F_f = \mu_s N$ for a body that is not sliding. That is only true at the very point of slipping. The memos give a mark for showing the inequality (needed vs maximum).

Worked example 12.2: will the block move?

`Final exam 2023, Q6 (Review lecture 2)`

A 25 kg block has a pulley on it. A cable pulled with $T = 40$ N at 30° above horizontal loops round the pulley and a fixed pulley, so the block feels two horizontal cable pulls of T plus the inclined pull. $\mu_s = 0.20$, $\mu_k = 0.15$. Find the block's acceleration.

1. Vertical ($a_y = 0$): $T\sin 30° + N - mg = 0$, so $N = 245.3 - 20 = 225.3$ N.

WHY: The upward part of the pull lifts some of the weight off the floor, which reduces friction.

2. Static check: friction needed to hold it is $2T + T\cos 30° = 114.6$ N. Maximum available is $\mu_s N = 45.05$ N. Needed > available, so **the block moves**.

3. Moving: $2T + T\cos 30° - \mu_k N = ma$, so $114.6 - 33.8 = 25a$.

a = 3.23 m/s² (memo: 3.23 m/s²)

Worked example 12.3: two bodies tied by a pulley `L20 Example 2`

Block A (60 kg) is on a 30° slope ($\mu_s = 0.25$, $\mu_k = 0.20$). Its cable runs over a fixed pulley and down to a movable pulley that carries cylinder B (20 kg). Released from rest, find both accelerations and the tension.

1. Constraint (Chapter 5): B hangs on a movable pulley, so $a_A = 2a_B$.

   WHY: The rope loops round B's pulley, so B moves half as far as A.

2. Static check: holding B needs $2T = m_B g$, so $T = 98.1$ N. A would then need $294.3 - 98.1 = 196.2$ N of friction, but only $\mu_s(509.7) = 127.4$ N is available. So A slides down the slope.

3. B, up positive: $2T - m_B g = m_B a_B$. A, down the slope positive: $m_A g \sin 30° - T - \mu_k N = m_A a_A$.

4. Substitute $a_A = 2a_B$ and solve the two equations together.

a_A = 1.45 m/s², a_B = 0.725 m/s², T = 105 N

Worked example 12.4: a force that depends on speed `L19 Example 1`

A 105 000 kg orbiter deploys its parachute at 320 km/h and drops it at 55 km/h. The drag is $F_D = 105v^2$ N. How far does it travel meanwhile?

1. Newton: $-105v^2 = 105\,000\,a$, so $a = -v^2/1000$.

2. a depends on v and we want a distance, so use $a\,ds = v\,dv$ (Chapter 2):

$$\int_0^s ds = \int_{88.9}^{15.3} -\frac{1000}{v}\, dv = 1000 \ln \frac{88.9}{15.3}$$

WHY: This is the Chapter 2 decision table again. Newton's law only supplies the formula for a.

s = 1761 m

IN ONE BREATH

- FBD (every force) = kinetic diagram ($m\vec{a}$). Choose axes along the acceleration.
- Friction: check "will it move?" with $\mu_s N$ first. If it slides, use $\mu_k N$ against the sliding.
- Connected bodies: one FBD per body, plus the rope constraint.
- Practice: L20 homework (M&K 3/20, 3/24, 3/28).

CHAPTER 13 · LECTURES L21 AND L24

Newton’s second law on curved paths

> In this chapter
>
> - $\sum F = ma$ in n–t and polar components
> - Why “centripetal force” is not a new kind of force
> - Banked tracks and losing contact over a hump
> - The rotating tube, and a memo slip to watch for

Nothing new in physics here: take the acceleration components from Chapter 3 and multiply by m.

$$\sum F_t = m\dot{v} \qquad \sum F_n = m\frac{v^2}{\rho} \qquad \sum F_r = m(\ddot{r} - r\dot{\theta}^2) \qquad \sum F_\theta = m(r\ddot{\theta} +$$

In 3D, cylindrical coordinates add $\sum F_z = m\ddot{z}$.

M V²/P IS NOT A FORCE

mv^2/ρ belongs on the *kinetic* diagram. It is what the real forces must add up to, in the direction of the centre of the bend. Something real has to provide it: friction on a flat corner, the normal force on a banked track, tension in a string, gravity for a satellite. If you draw it on the FBD as an extra force, you count it twice.

Worked example 13.1: a boat on a circle L21 Example 1

A 540 kg boat travels a circle of radius 40 m with speed $v = 0.0625t^2$ m/s. Find the normal and tangential forces from the water at $t = 10$ s.

1. Kinematics: $v = 6.25$ m/s, $a_t = \dot{v} = 0.125t = 1.25$ m/s², and $a_n = v^2/\rho = 0.977$ m/s².

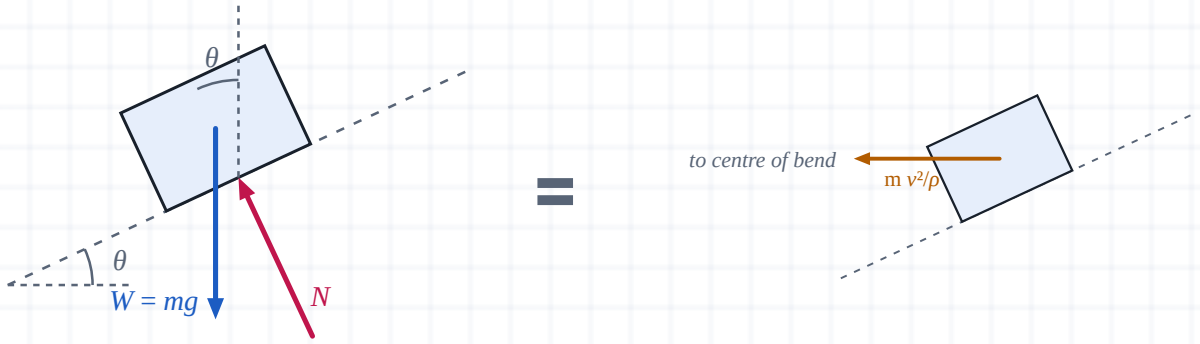
2. Multiply by the mass: $F_t = 540(1.25)$ and $F_n = 540(0.977)$.

   WHY: The water’s force is the only horizontal force, so it must supply both acceleration components.

F_n = 527 N (towards the centre), F_t = 675 N (forwards)

Worked example 13.2: a banked track with no friction needed L21 Example 3

At what angle θ should a track of radius ρ be banked so that a car at speed v needs no sideways friction? Then evaluate it for 90 km/h on a 100 m curve.



End-on view. The only forces are the weight and the normal force. Their sum must point horizontally to the centre of the bend.

1. Horizontal (towards the centre): $N\sin\theta = mv^2/\rho$.
2. Vertical (no vertical acceleration): $N\cos\theta - mg = 0$.
3. Divide the two equations to remove N and m: $\tan\theta = \dfrac{v^2}{\rho g}$.

   WHY: The tilted normal force has a sideways part. Banking makes the normal force do the friction's job.

4. For $v = 25$ m/s and $\rho = 100$ m: $\tan\theta = 625/981$.

$\theta = \tan^{-1}(v^2/\rho g) = 32.5°$ for 90 km/h on a 100 m curve

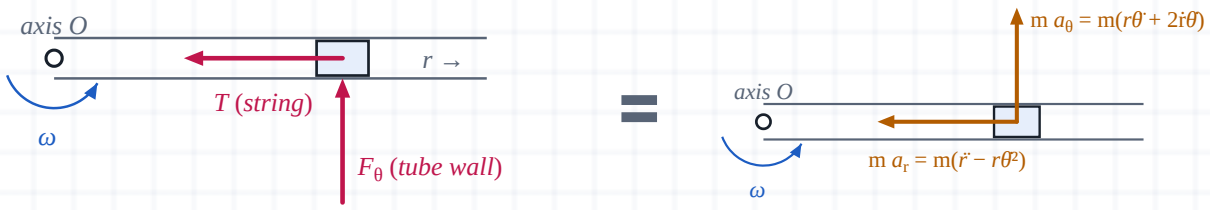
Losing contact over a hump

At the top of a hump of radius ρ, the centre of curvature is *below* the car. Taking down (towards the centre) as positive: $mg - N = mv^2/\rho$, so $N = m(g - v^2/\rho)$. The road can only push, so $N \geq 0$. The car leaves the road when $N = 0$, that is when $v = \sqrt{g\rho}$. L24 Example 1 (the Formula 1 car) is exactly this idea.

Polar coordinates: the rotating tube

Worked example 13.3: slider in a spinning tube `L21 Example 2`

A 0.2 kg slider moves in a horizontal tube that spins with $\dot\theta = 4$ rad/s and $\ddot\theta = -2$ rad/s². A string controls it: $r = 0.8$ m, $\dot r = -2$ m/s, $\ddot r = 4$ m/s². Find the string tension T and the side force F_θ from the tube.



Top view. The weight and the vertical tube reaction cancel, so they are left out of the horizontal picture.

1. List the five numbers: $r = 0.8$, $\dot r = -2$, $\ddot r = 4$, $\dot\theta = 4$, $\ddot\theta = -2$.

   WHY: Exactly as in Chapter 3: get these right and the rest is substitution.

2. Radial (outward positive; the string pulls inward):
   $-T = m(\ddot r - r\dot\theta^2) = 0.2(4 - 0.8 \times 16) = -1.76$ N.

3. Transverse: $F_\theta = m(r\ddot\theta + 2\dot r\dot\theta) = 0.2(-1.6 - 16) = -3.52$ N. The minus sign means the tube pushes on the slider from the other side.

T = 1.76 N, F_θ = 3.52 N (from the far side of the tube)

Worked example 13.4: the same tube in an exam

`Supp exam 2024, Q6 (Review lecture 2)`

$\omega = 5$ rad/s, $\dot\omega = -3$ rad/s², $m = 0.4$ kg, $r = 0.9$ m. The string is drawn down at 2.5 m/s but is decelerating at 4 m/s². Find the tension.

1. The five numbers: $\dot r = -2.5$ m/s (moving inwards), and $\ddot r = +4$ m/s² (the inward motion is slowing, so the radial acceleration is outward). $\dot\theta = \omega = 5$ rad/s.

2. $-T = m(\ddot r - r\dot\theta^2) = 0.4(4 - 0.9 \times 5^2) = 0.4(4 - 22.5)$.

T = 7.40 N

CHECK THIS ONE WITH YOUR LECTURER

The memo on the review slide writes $0.4\,(4 - 0.9(-3)^2)$ and gets 1.64 N. It has put $\dot{\omega} = -3$ where $\dot{\theta} = \omega = 5$ belongs. Using the same method that gives L21's 1.76 N, the tension is 7.40 N. Learn the method, which is the same either way, and ask in your next tutorial so you know what is expected.

IN ONE BREATH

- n–t: $\sum F_n = mv^2/\rho$ (towards the centre), $\sum F_t = m\dot{v}$.
- Polar: $\sum F_r = m(\ddot{r} - r\dot{\theta}^2)$, $\sum F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$.
- mv^2/ρ goes on the kinetic diagram only. A real force must provide it.
- Losing contact means $N = 0$.

CHAPTER 14 · LECTURES L25–L27

Work, energy and power

> In this chapter
>
> - What work is, and which forces do none
> - The work of weight and of a spring
> - $T_1 + \sum U = T_2$, and the potential-energy form $T_1 + V_1 + U' = T_2 + V_2$
> - Power and efficiency

THE BIG IDEA

Energy is bookkeeping. Kinetic energy is the account balance. Work is a deposit (positive) or a withdrawal (negative). Gravity and springs can be treated as savings accounts (potential energy) that you can draw back from. Because it is all scalars, you never need to resolve directions, and forces that do no work simply disappear from the problem.

Work

$$U_{1-2} = \int \vec{F} \cdot d\vec{r} = \int F \cos\alpha \, ds \qquad \text{(constant force: } U = F\cos\alpha \, \Delta s\text{)}$$

Only the part of the force **along** the motion does work. That gives three quick rules:

- Force along the motion: positive work. Force against it (friction, braking): negative work.
- **Zero work:** a force perpendicular to the motion (the normal force on a fixed surface), or a force at a point that does not move (a fixed pin, or the contact point of a wheel rolling without slip).
- Weight: $U_g = -mg\,\Delta y$. It is negative going up and positive going down, and only the height change matters, not the path.
- Linear spring: $U_s = \frac{1}{2}k(x_1^2 - x_2^2)$, where x is measured from the **unstretched** length.

COMMON SLIP

Using a change in length instead of the stretch. The spring formula needs the stretch at each end of the motion, measured from the spring's natural length. A spring that is already compressed at the start (Supp 2025, Chapter 16) starts with $x_1 \neq 0$.

The work–energy principle

$$T_1 + \sum U_{1-2} = T_2 \qquad \text{where} \qquad T = \tfrac{1}{2}mv^2$$

WHERE IT COMES FROM

Take the tangential part of Newton's law, $\sum F_t = m\,dv/dt$, and use the Chapter 2 trick $a\,ds = v\,dv$: $\sum F_t\,ds = mv\,dv$. Integrate along the path and the left side is the work, while the right side is $\tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2$. So it is Newton's law, integrated along the path, with time removed.

Move gravity and springs across to the energy side and you get the form L26 uses most:

$$T_1 + V_1 + U'_{1-2} = T_2 + V_2 \qquad V_g = mgy \text{ (from a datum you choose)} \qquad V_e = \tfrac{1}{2}kx^2$$

U' is the work of everything *except* gravity and springs (friction, applied forces, motors). If $U' = 0$, energy is conserved: $T_1 + V_1 = T_2 + V_2$.

COMMON SLIP

Counting gravity twice, once as U_g and again as V_g. Choose one bookkeeping style per problem.

Worked example 14.1: a collar dropping onto a spring

L25 Example 1

A 6 kg collar is released from rest and falls onto a spring ($k = 12$ kN/m). Find its speed when the spring is compressed 50 mm, given that the collar has then fallen 0.55 m in total.

1. States: 1 is the release ($T_1 = 0$), and 2 is at 50 mm compression.
   WHY: Energy compares two positions, so name them first.
2. Weight: $U_g = -mg\Delta y = -6(9.81)(-0.55) = +32.37$ J (it fell, so gravity did positive work).
3. Spring: $U_s = \tfrac{1}{2}(12\,000)(0^2 - 0.05^2) = -15.0$ J (the spring resists).
4. $0 + 32.37 - 15.0 = \tfrac{1}{2}(6)v^2$.

v = 2.41 m/s downwards

Power and efficiency (L27)

$$P = \frac{dU}{dt} = \vec{F}\cdot\vec{v} = Fv\cos\theta \qquad\qquad \varepsilon = \frac{P_{\text{out}}}{P_{\text{in}}} < 1$$

Power is how fast work is done, in watts (1 W = 1 J/s). Efficiency compares what you get out with what you put in. Friction inside the machine always takes a share, so ε is less than 1.

Worked example 14.2: winching a log `L27 Example 1`

A winch pulls a 360 kg log up a 30° slope at a constant 1.2 m/s. $\mu_k = 0.5$ and the winch efficiency is 0.80. Find the motor's power input.

1. Constant speed, so $a = 0$, and the forces balance (Chapter 12 with $ma = 0$).
2. $N = mg\cos 30^\circ = 3058$ N, so friction is $0.5N = 1529$ N.
3. Along the slope: $T = 1529 + mg\sin 30^\circ = 3295$ N.
4. $P_{\text{out}} = Tv = 3295 \times 1.2 = 3.95$ kW, and $P_{\text{in}} = P_{\text{out}}/0.8$.

P_{in} = 4.94 kW

A quick one in the same style (L27 Example 2): an escalator lifts 30 people per minute (65 kg each) up 7 m. Useful power is $\frac{30}{60}(65)(9.81)(7) = 2.23$ kW. With a 3 kW motor, $\varepsilon = 0.744$.

Note from Review lecture 2: questions on the energy of *rotating rigid bodies* (such as the flywheel bus) were not covered this year, so energy questions in your exam should involve particles.

IN ONE BREATH

- Work = force × distance along it. Normal forces, fixed pins and rolling contacts do no work.
- $T_1 + V_1 + U' = T_2 + V_2$, with $V_g = mgy$ and $V_e = \frac{1}{2}kx^2$ (x from the unstretched length).
- Use it for speed versus position. It cannot find forces perpendicular to the path.
- $P = Fv$, $\varepsilon = P_{\text{out}}/P_{\text{in}}$.

CHAPTER 15 · LECTURES L28–L29

Impulse and momentum: linear and angular

In this chapter

- Linear impulse and momentum, drawn as a vector triangle
- When momentum is conserved
- The two-stage pattern: momentum for the collision, energy afterwards
- Angular momentum about a fixed point

THE BIG IDEA

A force that acts for a while changes the momentum $m\vec{v}$. The total change is the force multiplied by the time it acts, the **impulse**. Use this when the question is about time, or about a hit so short that you could never know the force at each instant, only its average.

Linear impulse and momentum

$$m\vec{v}_1 + \sum \int_{t_1}^{t_2} \vec{F}\, dt = m\vec{v}_2$$

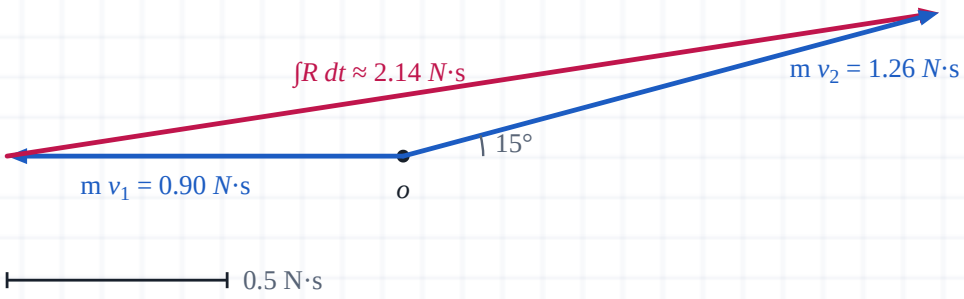
WHERE IT COMES FROM

$\sum \vec{F} = m\, d\vec{v}/dt$. Multiply by dt and integrate over time. The left side becomes the impulse, the right side the change in momentum. It is Newton's law integrated over time, just as work–energy was Newton's law integrated along the path.

- It is a **vector** equation: write it in x and y separately, or draw it as a triangle.
- Impulse is the area under the force–time graph. A constant force gives $\vec{F}\,\Delta t$.
- Units: N·s, which is the same as kg·m/s.

Worked example 15.1: a tennis shot `L28 Example 1`

A 60 g ball arrives horizontally at 15 m/s and leaves at 21 m/s, 15° above the horizontal, the other way. Contact lasts 0.02 s. Find the average force from the racket.



Momentum before plus impulse equals momentum after. The impulse closes the triangle (the weight's impulse, 0.012 N·s, is too small to see).

1. x (towards the leaving direction): $-0.06(15) + R_x(0.02) = 0.06(21\cos 15°)$, so $R_x = 105.9$ N.

   WHY: The ball reverses, so its starting momentum is negative. Signs carry the directions.

2. y (up): $0 + R_y(0.02) - 0.06(9.81)(0.02) = 0.06(21\sin 15°)$, so $R_y = 16.9$ N.

3. $R = \sqrt{105.9^2 + 16.9^2}$, at $\tan^{-1}(16.9/105.9)$.

R = 107 N at 9.07° above the horizontal

Conservation of momentum

For a *system* of bodies, the forces between them come in equal and opposite pairs (Newton's third law), so their impulses cancel. If the outside impulses are zero, or tiny compared with the impact forces, the total momentum is the same before and after:

$$\sum m\vec{v}_1 = \sum m\vec{v}_2$$

Weights and gentle ground forces are "non-impulsive": during a very short hit they hardly act at all, so you can ignore them.

Worked example 15.2: couple, then compress a spring `L28 Example 2`

Block B (10 kg, 15 m/s) hits block A (15 kg, at rest) on a smooth surface. They lock together and run into a spring ($k = 10$ kN/m). Find the maximum compression.

1. **Stage 1, the coupling** (momentum): $10(15) + 15(0) = 25v$, so $v = 6$ m/s.

WHY: The coupling forces are internal and enormous, and energy is lost as noise and heat. Momentum is the only thing you can trust here.

2. **Stage 2, the spring** (energy, nothing lost now): $\frac{1}{2}(25)(6^2) = \frac{1}{2}(10\,000)x^2$.
   WHY: Once the blocks move together, only the spring acts, so energy is conserved.

x = 0.300 m

This two-stage pattern (momentum through the collision, energy before and after it) is one of the most common in the exams.

Angular momentum

The angular momentum of a particle about a point O is the moment of its momentum:

$$\vec{H}_O = \vec{r} \times m\vec{v}, \quad |H_O| = mvd \qquad \sum M_O = \dot{H}_O \qquad H_{O1} + \sum \int M_O \, dt =$$

Here d is the perpendicular distance from O to the line of the velocity. If no external moment acts about O, H_O is conserved. That happens for a central force (always pointing at O), and for a body on a fixed pivot hit by something, because the pivot force passes through O.

Worked example 15.3: spinning up with a growing moment L29 Example 1

Four 5 kg spheres sit 0.6 m from a shaft on a light frame. A moment $M = 0.5t + 0.8$ *N·m is applied from rest. Find their speed after 4 s.*

1. Angular impulse: $\int_0^4 (0.5t + 0.8)\,dt = 0.25(16) + 0.8(4) = 7.2$ N·m·s.
2. Angular momentum after: $4 \times (0.6)(5)v = 12v$.
3. $0 + 7.2 = 12v$.

v = 0.600 m/s

Worked example 15.4: clay hits a pendulum `L29 Example 2`

A 12 g piece of clay at 2.5 m/s hits and sticks to a light bar 50 mm below the pivot.
An 80 g mass sits at the end of the 100 mm bar. Find ω just after impact.

1. About the pivot O: $H_{O1} = m_1 v_1 (L/2) = 0.012(2.5)(0.05) = 0.0015$ N·m·s.

   WHY: The pivot pushes hard during the impact, but that force has no moment about O. So angular momentum about O is conserved, while linear momentum is not.

2. After, both masses turn together: $H_{O2} = [m_1 (L/2)^2 + m_2 L^2]\,\omega = 0.00083\,\omega$.

ω = 1.81 rad/s

IN ONE BREATH

- $m\vec{v}_1 + \int \vec{F}\,dt = m\vec{v}_2$, a vector equation (use components or a triangle).
- No outside impulse means momentum is conserved. Use it for collisions and couplings.
- Momentum through the impact, energy before and after it.
- $H_O = mvd$. About a fixed pivot, the pivot force drops out.

CHAPTER 16 · LECTURE L30

Impact: collisions, bounces and the coefficient of restitution

In this chapter

- Central impact: momentum plus e
- Oblique impact: work along the line of impact and across it separately
- Energy lost in an impact
- A three-stage exam question: fall, impact, spring

THE BIG IDEA

A collision has two unknowns (the two speeds afterwards) but momentum conservation gives only one equation. The second equation comes from how "bouncy" the collision is, measured by the coefficient of restitution e.

Central impact

The **line of impact** runs through the two mass centres at the moment of contact. In central impact both bodies move along it.

$$m_A v_{A1} + m_B v_{B1} = m_A v_{A2} + m_B v_{B2} \qquad e = \frac{v_{B2} - v_{A2}}{v_{A1} - v_{B1}} = \frac{\text{speed of separation}}{\text{speed of approach}}$$

- $e = 1$: perfectly elastic, no energy lost (an ideal that real collisions never quite reach).
- $e = 0$: perfectly plastic. The bodies stick together and the most energy possible is lost.
- Typical values: steel 0.5–0.8, wood 0.4–0.6, glass about 0.94.

WHY E USES RELATIVE SPEEDS

How hard two bodies squash together depends only on how fast they close on each other, not on how fast either one moves over the ground. So the fair measure of bounciness compares the closing speed before with the opening speed after. Keep one direction positive for *every* velocity, and the signs take care of themselves.

Worked example 16.1: a quick central impact `L30 Think-Pair-Share 1`

Block B (1 kg, 10 m/s →) squarely hits block A (3 kg, at rest). Afterwards A moves at 4 m/s →. Find v_{B2} and e.

1. Momentum (→ positive): $1(10) + 0 = 1\,v_{B2} + 3(4)$, so $v_{B2} = -2$ m/s.
2. $e = \dfrac{4 - (-2)}{10 - 0} = 0.6$.

   WHY: B was behind, so the separation speed is A's velocity minus B's.

v_{B2} = 2 m/s ←, e = 0.6

Worked example 16.2: a bouncing ball `L30 Example 1`

A ball thrown horizontally at 2.4 m/s from 0.9 m high bounces off hard ground with $e = 0.8$. Find the rebound speed and the height it reaches.

1. Falling (Chapter 3 projectile): $v_y^2 = 2(9.81)(0.9)$, so $v_y = 4.20$ m/s down.
2. Impact with the fixed ground: only the part along the line of impact (vertical) changes, $v_y' = e\,v_y = 3.36$ m/s up. The horizontal 2.4 m/s is unchanged on a smooth surface.
3. Rebound speed $\sqrt{2.4^2 + 3.36^2}$, and height $h = 3.36^2/(2 \times 9.81)$.

v = 4.13 m/s, h = 0.576 m

Oblique impact

Now the velocities are at an angle to the line of impact. Set up an n axis along the line of impact and a t axis across it (the contact plane). Then:

- **Along n:** the momentum of the *system* is conserved, and e applies, exactly as in central impact.
- **Along t:** with smooth surfaces there is no force across the contact, so *each* body keeps its own t-velocity.

That gives four equations for the four unknowns (two components for each body).

Worked example 16.3: two hockey pucks `L30 Example 2`

Identical pucks: A moves at 6 m/s along the line of impact (x). B moves at 10 m/s, with $-10\cos 30°$ along x and $+5$ m/s across it. $e = 0.75$.

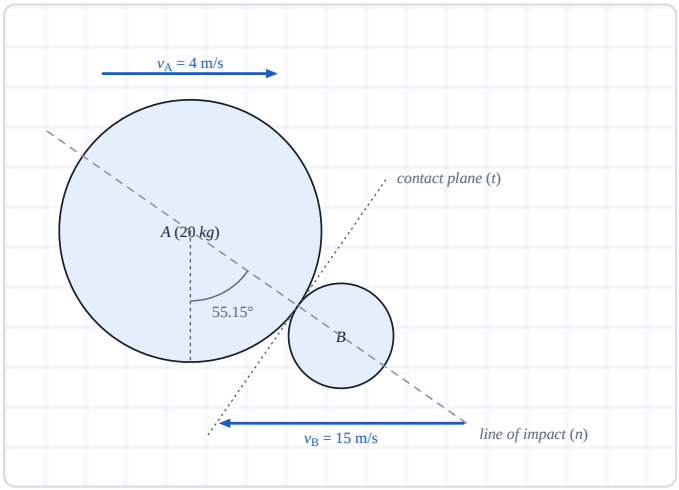
1. x-momentum (equal masses): $6 - 8.66 = v_{Ax2} + v_{Bx2}$.
2. Restitution: $v_{Bx2} - v_{Ax2} = 0.75(6 + 8.66) = 11.0$.
3. Solve: $v_{Ax2} = -6.83$ m/s and $v_{Bx2} = 4.17$ m/s.
4. y: A keeps 0 and B keeps 5 m/s, so $v_B = \sqrt{4.17^2 + 5^2}$ at $\tan^{-1}(5/4.17)$.

v_A = 6.83 m/s ←, v_B = 6.51 m/s at 50.2° to the x-axis

Worked example 16.4: two spheres, from a supp exam

`Supp exam Q7 (Review lecture 1)`

Sphere A (20 kg) moves → at 4 m/s. Sphere B (5 kg) moves ← at 15 m/s on a parallel path. $e = 0.6$, and the line of impact is at 55.15° to the vertical. Find A's velocity just after impact, relative to the line of impact.





`Left: geometry. Right: A's velocity split along and across the line of impact.`

1. Components along n (from A towards B): $u_A = 4\sin 55.15° = 3.283$ and $u_B = -15\sin 55.15° = -12.31$ m/s. Across n: A has $4\cos 55.15° = 2.286$ m/s, and keeps it.
2. Momentum along n: $20(3.283) + 5(-12.31) = 20v_{A\parallel} + 5v_{B\parallel}$.

3. Restitution: $v_{B\parallel} - v_{A\parallel} = 0.6(3.283 + 12.31) = 9.355$.
4. Solve: $v_{A\parallel} = -1.707$ m/s (A bounces back along the line). Then $v_A = \sqrt{1.707^2 + 2.286^2}$, at $\tan^{-1}(2.286/1.707)$ to the line of impact.

v_A = 2.85 m/s at 53.3° to the line of impact

CHECK THIS ONE WITH YOUR LECTURER

The review-slide memo uses exactly these equations but reports $(v_{A2})_{\parallel} = -1.855$ m/s, giving 2.94 m/s at 50.9°. Putting −1.855 back into its own restitution equation gives $e = 0.65$, not 0.6. The value −1.707 satisfies both momentum and $e = 0.6$. The method is what earns most of the marks, and it is identical either way.

Energy lost in an impact

Momentum survives a collision. Kinetic energy usually does not: $\text{loss} = \sum T_1 - \sum T_2$, and it goes into heat, sound and permanent dents. So never use energy conservation *across* an impact unless you are told it is perfectly elastic.

Worked example 16.5: putty, block and springs

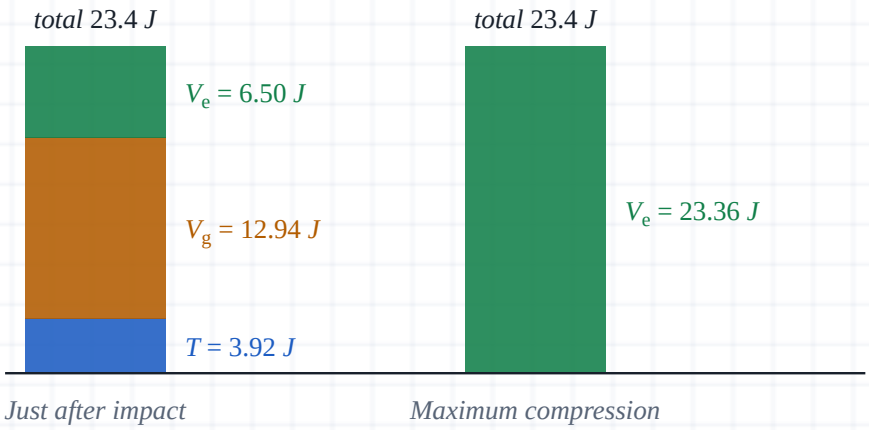
`Supp exam 2025, Q5 (Review lecture 2)`

2 kg of putty is dropped 2 m onto an 18 kg block resting on two springs ($k = 1.2$ kN/m each). It sticks. Find the maximum additional deflection δ.

1. **Stage 1, the fall** (energy): $v_P = \sqrt{2(9.81)(2)} = 6.264$ m/s.
2. **Stage 2, the impact** (momentum, plastic): $2(6.264) = 20v_3$, so $v_3 = 0.6264$ m/s.

   WHY: This is the lossy step: 39.2 J arrives, and only 3.9 J survives as kinetic energy.
3. **Before impact the springs are already compressed** by the block: $2kx_0 = 18g$, so $x_0 = 0.0736$ m.

   WHY: The springs' starting energy is not zero. Missing this is the most common way to lose marks on this question.
4. **Stage 3, the compression** (energy, datum at the lowest point): $T_3 + V_{g3} + V_{e3} = V_{e4}$, which is $3.924 + 196.2\,\delta + 6.496 = 1200(\delta + 0.0736)^2$.
5. Solve the quadratic and keep the positive root.



Energy bookkeeping for stage 3. The total stays at 23.4 J. Kinetic and gravitational energy are both handed to the springs.

δ = 65.9 mm (memo: 65.9 mm)

IN ONE BREATH

- Along the line of impact: system momentum plus e = separation ÷ approach.
- Across it (smooth): each body keeps its own velocity component.
- Energy is lost in impacts ($e < 1$), so switch to momentum for the impact and back to energy afterwards.

CHAPTER 17 · LECTURE L32 (MOI)

Mass moment of inertia: rotational mass

In this chapter

- What I measures, and why distance is squared
- The standard shapes you need
- Radius of gyration
- The parallel-axis theorem and composite bodies

THE BIG IDEA

Mass resists being *pushed*: $F = ma$. The moment of inertia resists being *spun*: $M = I\alpha$. It depends not just on how much mass there is, but on how far it sits from the axis. Mass far from the axis counts for a lot more.

$$I = \int r^2 \, dm \qquad \text{units: } \mathrm{kg \cdot m^2}$$

WHY R IS SQUARED

A bit of mass at radius r has to move at $r\omega$, so it takes more effort to speed up. Its tangential acceleration $r\alpha$ also gives it more leverage against the turning, because the moment arm is another r. One r from each effect gives r^2. That is why a figure skater spins faster when pulling the arms in (Chapter 15: angular momentum $I\omega$ stays constant while I drops).

L32 says plainly that integrating to find I will **no longer** come up in tests or exams. You will use the table (M&K Table D/4), the parallel-axis theorem, and addition.

The shapes you need

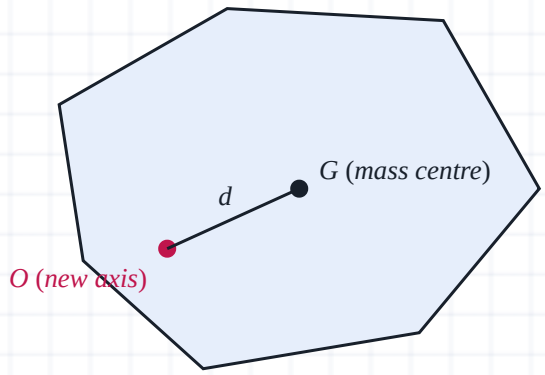
BODY (MASS m)	AXIS	I
Slender rod, length L	Through the centre, ⊥ rod	$\frac{1}{12}mL^2$
Slender rod, length L	Through one end, ⊥ rod	$\frac{1}{3}mL^2$
Thin ring or hoop, radius r	Through the centre, ⊥ plane	mr^2
Solid disc or cylinder, radius r	Its own axis	$\frac{1}{2}mr^2$
Thin disc, radius r	A diameter	$\frac{1}{4}mr^2$
Solid sphere, radius r	A diameter	$\frac{2}{5}mr^2$
Thin rectangular plate $a \times b$	Through the centre, ⊥ plate	$\frac{1}{12}m(a^2+b^2)$

Sanity check: the hoop (mr^2) beats the disc ($\frac{1}{2}mr^2$) because all of the hoop's mass is at the rim.

Radius of gyration

$I = mk^2$, so $k = \sqrt{I/m}$. It is the radius at which you could squash all the mass into a thin ring without changing I. Questions often give k instead of I (the I-beam in Chapter 19 has $k_O = 2.32$ m).

The parallel-axis theorem



Moving the axis a distance d away from the mass centre always increases I, by md^2.

$$I_O = I_G + md^2$$

COMMON SLIP

One end of the move must be the mass centre G. To go from axis O to axis P, go O → G (subtract md^2_{OG}), then G → P (add md^2_{GP}).

Worked example 17.1: a rod about a quarter point

L32 Think-Pair-Share 3

Find I for a slender rod (mass M, length L) about an axis through the point $L/4$ from one end.

1. The axis is $d = L/4$ from the centre, so $I = \frac{1}{12}ML^2 + M(L/4)^2 = ML^2\left(\frac{4}{48} + \frac{3}{48}\right)$.

$I = 7ML^2/48$

Worked example 17.2: a composite pendulum

In the style of L32 Example 1

A pendulum is a 3 kg slender rod (0.8 m), pivoted at its top end O, with a 5 kg solid disc (radius 0.1 m) centred on its bottom end. Find I_O, the position of G, and I_G.

1. Rod about O (its end): $\frac{1}{3}(3)(0.8)^2 = 0.640$.
2. Disc about O: its own $\frac{1}{2}(5)(0.1)^2 = 0.025$, plus $5(0.8)^2 = 3.200$ from the parallel-axis theorem.

   WHY: The disc's own spin barely matters. Almost all of its I comes from being far from O.
3. $I_O = 0.640 + 0.025 + 3.200 = 3.865$ kg·m².
4. Mass centre: $\bar{y} = \dfrac{3(0.4) + 5(0.8)}{8} = 0.650$ m below O. Then $I_G = I_O - 8(0.65)^2$.

I_O = 3.87 kg·m², G is 0.650 m below O, I_G = 0.485 kg·m²

IN ONE BREATH

- I is rotational mass, and distance from the axis counts squared.
- Know the table, and $I = mk^2$.
- $I_O = I_G + md^2$, always to or from G.
- Composites: add each part about the same axis.

CHAPTER 18 · LECTURES L33–L34 (RECTILINEAR TRANSLATION)

Rigid bodies that slide without turning

> In this chapter
>
> - The three equations of planar rigid-body motion
> - Why a translating body still needs a moment equation
> - Taking moments about a smart point, $\sum M_P = ma_G d$
> - The boom on the bakkie (review lecture)

A rigid body can rotate as well as move, so it needs one more equation than a particle. For planar motion there are three:

$$\sum F_x = m(a_G)_x \qquad \sum F_y = m(a_G)_y \qquad \sum M_G = I_G\,\alpha$$

The force equations use the acceleration of the **mass centre G**. The moment equation is taken **about G**, with I meaning I_G (L33 marks this “NB!”).

There is also a very useful alternative. Take moments about *any* point P, and include the moment of the kinetic diagram on the right-hand side:

$$\sum M_P = I_G\,\alpha + ma_G\,d$$

Here d is the perpendicular distance from P to the line of $m\vec{a}_G$.

WHY THE EXTRA TERM

The forces on the FBD and the $m\vec{a}_G$ (plus $I_G\alpha$) on the kinetic diagram are equivalent, so they must have the same moment about every point, not just about G. About G, $m\vec{a}_G$ has no moment, which leaves $I_G\alpha$. About any other point it does, $ma_G d$. Pick P where two unknown forces meet, and both drop out of the equation at once.

Translation: no rotation, but moments still matter

In translation, $\omega = \alpha = 0$, so $\sum M_G = 0$. That does *not* make the moment equation useless. It is what decides how the load is shared between wheels, pins and cables. About any other point P, $\sum M_P = ma_G d$.

Worked example 18.1: a wheelie `L33 Think-Pair-Share 1`

A motorbike and rider (295 kg) hold a wheelie while accelerating at 5 m/s². Find the friction force on the back wheel.

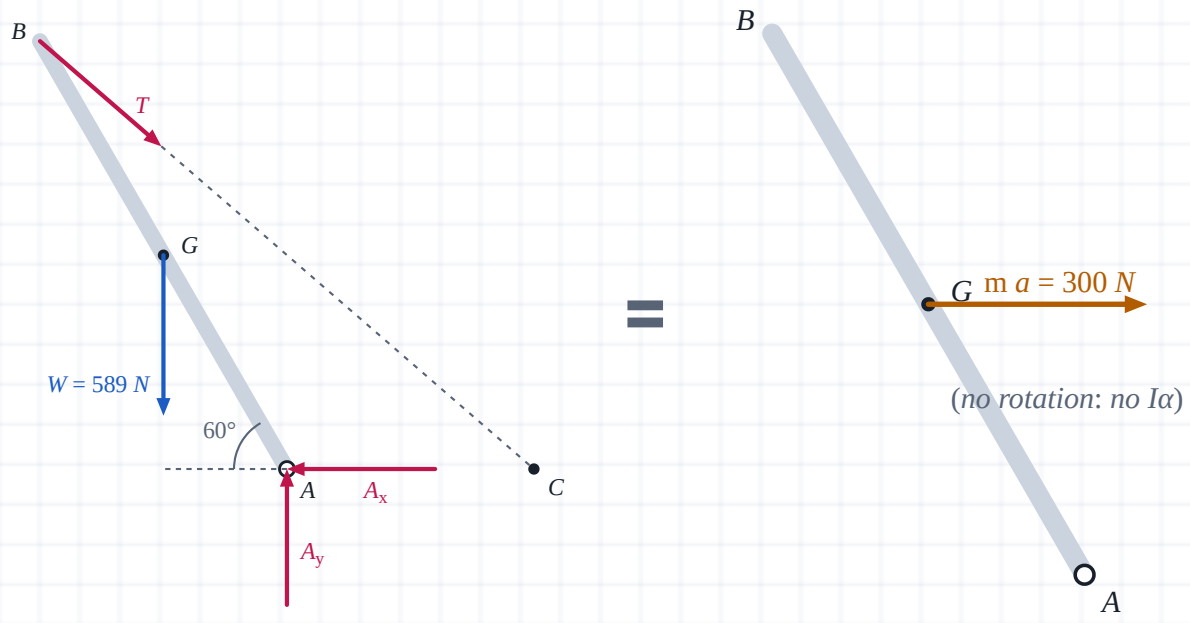
1. The front wheel is off the ground, so the back wheel's friction is the only horizontal force: $F = ma = 295 \times 5$.

   WHY: The moment equation about G is what holds the wheelie at a steady angle. The horizontal force equation alone answers this question.

F = 1475 N

Worked example 18.2: the boom on the bakkie `Review lecture 1, Example 2`

A uniform 4 m, 60 kg boom is pinned at A to the back of a bakkie, at 60° to the horizontal, and held by a cable from its top B to C, 2 m in front of A. The bakkie accelerates forwards at 5 m/s². Find the cable tension and the total pin force at A.



The boom moves with the bakkie without turning, so the kinetic diagram is just $m\vec{a}$ at G.

1. Geometry: B is at $(-2, 3.464)$ m from A. The cable angle at C is $\theta = \tan^{-1}\dfrac{4\sin 60°}{4\cos 60° + 2} = 40.9°$. G is 1.0 m behind and 1.732 m above A.

2. Moments about A, clockwise positive. This removes A_x and A_y in one go:

$$T(2\sin 40.9^\circ) - 589(1.0) = \underbrace{300 \times 1.732}_{ma_G d}$$

so $T = 846$ N.

WHY: The cable's moment arm about A is the perpendicular distance from A to the line BC, which is $AC\sin\theta$. The kinetic diagram's moment is ma times the height of G above A.

3. Horizontal: $T\cos 40.9^\circ - A_x = 300$, so $A_x = 340$ N (acting backwards on the boom).

4. Vertical (no vertical acceleration): $A_y - 589 - T\sin 40.9^\circ = 0$, so $A_y = 1143$ N.

T = 846 N, A = √(340² + 1143²) = 1192 N (memo: 846 N and 1192 N)

IN ONE BREATH

- $\sum F = ma_G$ and $\sum M_G = I_G\alpha$, or $\sum M_P = I_G\alpha + ma_G d$ about any P.
- Translation: $\alpha = 0$, but moments still decide the reactions.
- Take moments where the unknowns meet.
- Practice: M&K 6/10 in L33 (T = 24.5 N, F_A = 32.9 N).

CHAPTER 19 · LECTURE L32 (RIGID-BODY ROTATION I–II)

Rigid bodies spinning about a fixed pivot

> In this chapter
>
> - The equations in n–t form, using Chapter 6's kinematics
> - Why $\sum M_O = I_O\alpha$ works about the pivot
> - "An instant after release": what that phrase gives you
> - Two worked examples, including the I-beam from L32

Every point of the body moves on a circle around the fixed axis O, including the mass centre G at radius $\bar{r}$. So G's acceleration has the two parts from Chapter 6, and the force equations are best written in n–t components:

$$\sum F_n = m\bar{r}\omega^2 \text{ (towards } O) \qquad \sum F_t = m\bar{r}\alpha \qquad \sum M_O = I_O\,\alpha$$

WHY MOMENTS ABOUT THE PIVOT USE I_O

Take moments about O with the Chapter 18 rule: $\sum M_O = I_G\alpha + ma_t\bar{r}$. The tangential part $m\bar{r}\alpha$ has moment arm $\bar{r}$, and the normal part points straight at O, so it has no moment. That gives $\sum M_O = (I_G + m\bar{r}^2)\alpha = I_O\alpha$, by the parallel-axis theorem (Chapter 17). The moment equation about the pivot is the most useful one, because the unknown pin forces at O drop out.

Procedure (L32)

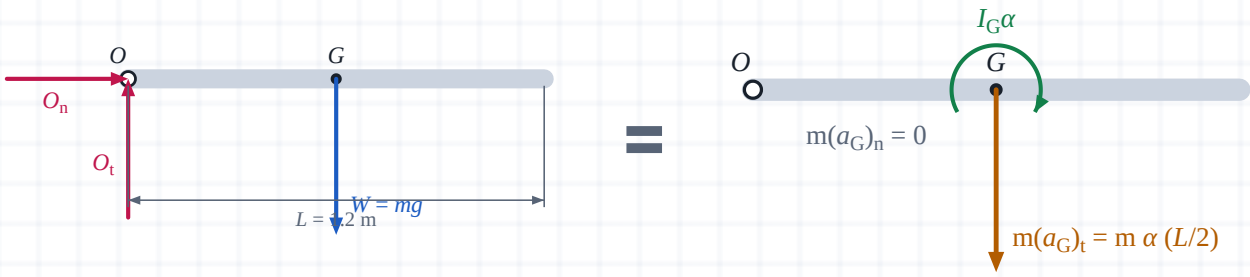
1. Use n–t axes centred on O.
2. Draw the FBD: weight, applied loads and the two pin components O_n, O_t.
3. Draw the kinetic diagram: $m\bar{r}\omega^2$ towards O, $m\bar{r}\alpha \perp$ OG, and $I_G\alpha$.
4. Find α from $\sum M_O = I_O\alpha$ first, then the pin forces from the two force equations.

"AN INSTANT AFTER RELEASE FROM REST"

This phrase means $\omega = 0$, so every ω^2 term is zero. But α is *not* zero: the body has just started to speed up. Every exam uses this phrase. It makes the normal equation trivial and leaves you with α and the tangential force.

Worked example 19.1: a rod released from horizontal `The classic case`

A uniform 4 kg rod, 1.2 m long, is pinned at one end and released from rest in the horizontal position. Find its initial angular acceleration and the pin force.



At release $\omega = 0$, so the kinetic diagram has only the tangential term and $I_G\alpha$.

1. $I_O = \frac{1}{3}mL^2 = 1.92$ kg·m².
2. $\sum M_O = I_O\alpha$: $mg(L/2) = 1.92\,\alpha$, so $\alpha = \dfrac{3g}{2L} = 12.3$ rad/s² clockwise.
3. Normal: $O_n = m\bar{r}\omega^2 = 0$.
4. Tangential (down positive): $mg - O_t = m\bar{r}\alpha = 4(0.6)(12.26)$, so $O_t = 39.24 - 29.43$.

   WHY: G drops at $\frac{3}{4}g$, so the pin only needs to hold a quarter of the weight. The pin force suddenly falls from mg to $mg/4$ at the moment of release.

α = 12.3 rad/s² CW, pin force = 9.81 N (= mg/4) upwards

Worked example 19.2: the I-beam on a bracket `L32 Rotation II, Example 2`

A 300 kg beam with $k_O = 2.32$ m is pinned at O on a small bracket. G is 1.8 m along and 0.9 m across from O. The support at B is removed. Find the pin force at O at that instant.

1. $I_O = mk_O^2 = 300(2.32)^2 = 1615$ kg·m².
2. $\sum M_O = I_O\alpha$: the weight's moment arm is the horizontal distance 1.8 m, so $300(9.81)(1.8) = 1615\,\alpha$ and $\alpha = 3.28$ rad/s² CW.
3. Geometry of OG: $\bar{r} = \sqrt{1.8^2 + 0.9^2} = 2.012$ m, at $\beta = \tan^{-1}(0.9/1.8) = 26.6°$.
4. Tangential (⊥ OG): $mg\cos\beta - F_{O,t} = m\bar{r}\alpha$, so $F_{O,t} = 2632 - 1981 = 651$ N.

5 Normal (along OG, $\omega = 0$): $F_{O,n} - mg\sin\beta = 0$, so $F_{O,n} = 1316$ N.

WHY: Resolve the weight into the n and t directions, because those are the directions of G's acceleration.

$F_O = \sqrt{(651^2 + 1316^2)} = 1469$ N (memo: 1469 N)

IN ONE BREATH

- G moves on a circle: $\sum F_n = m\bar{r}\omega^2$, $\sum F_t = m\bar{r}\alpha$.
- $\sum M_O = I_O\alpha$, with $I_O = I_G + m\bar{r}^2$. Use it first, because the pin forces drop out.
- "Instant after release": $\omega = 0$, $\alpha \neq 0$.
- Practice: L32 homework (M&K 6/65, 6/36).

CHAPTER 20 · LECTURES L33–L35 (PLANE MOTION I–III)

Rigid bodies that move and turn at once

> In this chapter
>
> - Where Part I and Part II meet: linking a_G and α
> - Rolling: assume no slip, then check
> - Slipping: friction becomes $\mu_k N$
> - Three worked examples, including the rod-and-hoop review question

Same three equations as Chapter 18, now with $\alpha \neq 0$:

$$\sum \vec{F} = m\vec{a}_G \qquad \sum M_G = I_G\alpha \qquad \sum M_P = I_G\alpha + ma_G d$$

THE BIG IDEA

Three equations, but often four unknowns: two force components, a_G and α. The fourth equation is **kinematics**, which links $\vec{a}_G$ to α. For rolling without slip it is $a_G = \alpha r$. For a pinned or sliding end it is the relative acceleration equation from Chapter 10, $\vec{a}_G = \vec{a}_A + (\vec{a}_{G/A})_n + (\vec{a}_{G/A})_t$. This is why Part I matters so much for Part II.

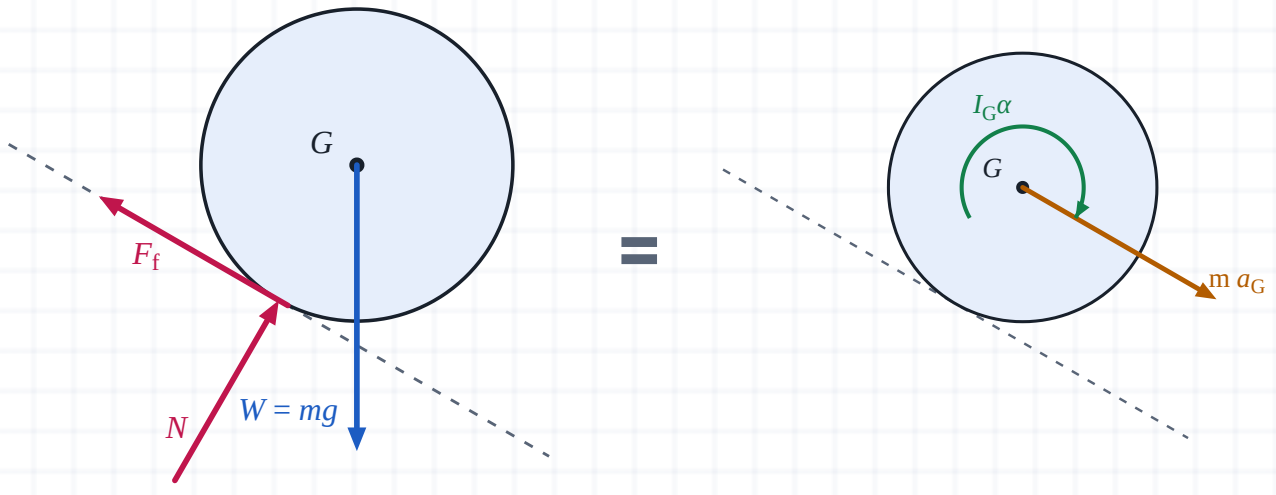
Rolling bodies (L33): assume, solve, check

1. Draw the FBD (weight, normal force, friction) and the kinetic diagram ($m\vec{a}_G$, $I_G\alpha$).
2. **Assume pure rolling**: $a_G = \alpha r$. Friction is unknown, anything up to $\mu_s N$.

   WHY: Static friction adjusts itself, so you cannot write $F = \mu_s N$. Treat it as an unknown.
3. Solve. Moments about the contact point are fastest, because F and N both pass through it.
4. **Check**: is the friction needed $\leq \mu_s N$? If yes, the body rolls and you are done.
5. If not, it **slips**: now $F = \mu_k N$ (opposing the slip), and $a_G \neq \alpha r$. Solve the force and moment equations separately.

Worked example 20.1: a cylinder rolling down a ramp L33 Example 2

A solid cylinder is released from rest on a 30° ramp with $\mu_s = 0.30$, $\mu_k = 0.20$.
Find a_G and the friction force, and check that it rolls.



Friction must point up the slope: it supplies the clockwise moment that makes the cylinder spin.

1. Moments about the contact point C (both unknown contact forces drop out):

$$mg\sin 30^\circ\, r = I_G\alpha + ma_G r = \tfrac{1}{2}mr^2\alpha + m(\alpha r)r$$

so $\alpha = \dfrac{2g\sin 30^\circ}{3r}$ and $a_G = \alpha r = \tfrac{2}{3}g\sin 30^\circ$.

2. Along the slope: $mg\sin 30^\circ - F = ma_G$, so $F = \tfrac{1}{3}mg\sin 30^\circ$.

3. Check: friction needed ÷ N is $\tfrac{1}{3}\tan 30^\circ = 0.19$, which is less than $\mu_s = 0.30$, so it rolls.

WHY: Neither m nor r appears in the answer. Every solid cylinder rolls down this ramp in the same way.

a_G = 3.27 m/s² down the ramp; F = ⅓mg sin 30° up the ramp (5.89 N for the slide's cylinder, below its 9.17 N limit)

Worked example 20.2: a wheel that slips L34 Example 1

A 10 kg wheel (radius 200 mm, $k_G = 180$ mm) is released on a 60° incline.
$\mu_s = 0.35$, $\mu_k = 0.30$. Find α.

1. Try rolling: moments about the contact point give $a_G = \dfrac{g\sin 60^\circ}{1 + k^2/r^2} = 4.69$ m/s². Then $F = mg\sin 60^\circ - ma_G = 38.0$ N.

2. Check: $N = mg\cos 60^\circ = 49.05$ N, so at most $\mu_s N = 17.2$ N is available. 38.0 > 17.2, so **it slips**.

3. Slipping: $F = \mu_k N = 14.7$ N. Moments about G: $Fr = mk^2\alpha$, so $\alpha = \dfrac{14.7(0.2)}{10(0.18)^2}$.

   WHY: Once it slips, rolling no longer ties a_G to α. Friction alone sets the spin.

α = 9.08 rad/s² (and a_G = 7.02 m/s² down the slope)

Worked example 20.3: the rod on a hoop

`Review lecture 1`

A uniform rod ($l = 250$ mm, $m = 0.5$ kg) lies along a horizontal radius of a light hoop of radius l, so G is $l/2$ from the hoop's centre O. The assembly is released from rest on a horizontal surface and rolls. Find the initial α.

1. **Kinematics (Chapter 10):** $\vec{a}_G = \vec{a}_O + (\vec{a}_{G/O})_n + (\vec{a}_{G/O})_t$. Rolling gives $a_O = l\alpha$ (horizontal). From rest, $\omega = 0$, so the normal term vanishes and $(a_{G/O})_t = \frac{l}{2}\alpha$ (vertical, downwards).

   WHY: This is the step that needs Part I. It tells you both components of $\vec{a}_G$ in terms of the single unknown α.

2. Horizontal: $F_f = m\,l\alpha$.

3. Vertical: $N - mg = -m\frac{l}{2}\alpha$.

4. Moments about G (CCW positive): $-N\frac{l}{2} + F_f\,l = -\frac{1}{12}ml^2\alpha$.

5. Substitute and simplify: $\alpha\,l\left(\frac{1}{4} + 1 + \frac{1}{12}\right) = \frac{g}{2}$, so $\alpha = \dfrac{3g}{8l}$.

α = 14.7 rad/s² (memo: 14.7 rad/s²)

Free bodies. A body that is not attached to anything (L33 rocket, L34 Example 2) needs no kinematic link: $\sum F = ma_G$ gives a_G, and $\sum M_G = I_G\alpha$ gives α. In L34 Example 2, a 6 N push at one end of a 1.2 kg body gives $a_G = 5$ m/s² and $\alpha = 48.8$ rad/s². The pushed end then accelerates at $5 + 0.425(48.8) = 25.7$ m/s². Pushing off-centre makes things spin as well as move.

Linkages (L35). L35 Example 1 (a crank driving a bar on a roller) is the full combination: a Chapter 9 velocity diagram for ω_{AB}, a Chapter 10 acceleration diagram for $\vec{a}_G$ and α, then this chapter's three equations for the roller force. Work through it once you are comfortable with both halves.

COMMON SLIPS

- Writing $F = \mu_s N$ for a body that rolls. It is an unknown, up to $\mu_s N$.
- Using $a_G = \alpha r$ after finding that the body slips.
- Taking moments about the contact point with $\sum M = I_G \alpha$ alone. About any point other than G you need the $m a_G d$ term too (for a round body rolling on a flat surface this adds up to $I_C \alpha$, because $m a_G r = m r^2 \alpha$).
- Forgetting the kinematic link, which leaves you one equation short.

IN ONE BREATH

- $\sum F = m a_G$, $\sum M_G = I_G \alpha$, plus a kinematic link between a_G and α.
- Rolling: assume $a_G = \alpha r$, solve, check $F \leq \mu_s N$. If it fails, $F = \mu_k N$ and the link is gone.
- Moments about the contact point remove both contact forces.

Kinetics playbook and one-page formula sheet

The kinetics question checklist

1. **Read for the tool.** Instant, force or acceleration: Newton. Speed against position: energy. Speed against time, or a short hit: impulse. Collision: momentum plus e. (See the table at the start of Part II.)
2. **Split the story into stages** wherever something sudden happens (an impact, a cable going slack, contact being lost). Each stage may need a different tool.
3. **Draw the FBD and the kinetic diagram** for each body. Label every force. Mark the positive directions.
4. **Friction:** check whether it moves, or rolls, before choosing μ_s or μ_k.
5. **Count unknowns against equations.** If you are short, the missing equation is kinematics: a rope constraint, $a_G = \alpha r$, or a relative acceleration.
6. **Rigid bodies:** take moments about the point where the most unknowns meet (a pin, or the contact point), including $ma_G d$ if that point is not G.
7. **Answer with direction and units**, for example “846 N”, “3.27 m/s² down the incline”, “14.7 rad/s² CW”.

Formula sheet: Part II on one page

TOPIC	FORMULAS	REMEMBER
Newton (particle)	$\sum F_x = ma_x,\ \sum F_y = ma_y$ $\sum F_t = m\dot{v},\ \sum F_n = mv^2/\rho$ $\sum F_r = m(\ddot{r} - r\dot{\theta}^2),\ \sum F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$	$m\vec{a}$ goes on the kinetic diagram, never on the FBD.
Friction	Static: $F \le \mu_s N$. Sliding: $F = \mu_k N$.	Check whether it moves first.
Work	$U = \int F\cos\alpha\, ds$, $U_g = -mg\Delta y$, $U_s = \frac{1}{2}k(x_1^2 - x_2^2)$	x from the unstretched length. Normal forces and fixed pins do no work.
Energy	$T_1 + V_1 + U'_{1-2} = T_2 + V_2$, $T = \frac{1}{2}mv^2$, $V_g = mgy$, $V_e = \frac{1}{2}kx^2$	Scalar. Not valid across an impact unless $e = 1$.
Power	$P = \vec{F}\cdot\vec{v}$, $\varepsilon = P_{\text{out}}/P_{\text{in}}$	1 W = 1 J/s.
Linear momentum	$m\vec{v}_1 + \sum\int \vec{F}\,dt = m\vec{v}_2$	Vector. Use components or a triangle.
Angular momentum	$H_O = mvd$, $\sum M_O = \dot{H}_O$, $H_{O1} + \int M_O\,dt = H_{O2}$	About a fixed pivot, the pivot force drops out.
Impact	Along n: $\sum mv$ conserved, $e = \dfrac{v_{B2} - v_{A2}}{v_{A1} - v_{B1}}$. Along t: each body keeps its v_t.	$e = 1$ elastic, $e = 0$ they stick.
Moment of inertia	Rod $\frac{1}{12}mL^2$ (centre), $\frac{1}{3}mL^2$ (end). Disc $\frac{1}{2}mr^2$. Hoop mr^2. Sphere $\frac{2}{5}mr^2$. $I = mk^2$, $I_O = I_G + md^2$	Parallel axis always to or from G.
Rigid body	$\sum\vec{F} = m\vec{a}_G$, $\sum M_G = I_G\alpha$, $\sum M_P = I_G\alpha + ma_G d$	Translation: $\alpha = 0$.
Fixed axis	$\sum F_n = m\bar{r}\omega^2$, $\sum F_t = m\bar{r}\alpha$, $\sum M_O = I_O\alpha$	Released from rest: $\omega = 0$, $\alpha \neq 0$.
Rolling	No slip: $a_G = \alpha r$, $F \le \mu_s N$. Slipping: $F = \mu_k N$, no link.	Assume rolling, then check.

Two memo answers to ask about

While checking every example for this book, two review-lecture memo answers did not reproduce. The methods in the memos are right. Only a number slipped.

- **Supp 2024 Q6 (rotating tube):** the memo uses $\dot{\omega} = -3$ where $\dot{\theta} = 5$ belongs, and gets $T = 1.64$ N. With $\dot{\theta} = 5$, $T = 7.40$ N (Chapter 13).
- **Supp Q7 (two spheres):** the memo's $v_{A\parallel} = -1.855$ m/s gives $e = 0.65$ in its own equation. With $e = 0.6$, $v_{A\parallel} = -1.707$ m/s and $v_A = 2.85$ m/s at 53.3° (Chapter 16).

Also, from L16 in Part I: the slide uses a rod length of 0.250 m in the normal-acceleration step where the real length is 0.350 m (its answer, 305 m/s², is correct).

THE WHOLE OF PART II IN FOUR SENTENCES

Draw every force on the body, and $m\vec{a}$ (plus $I_G\alpha$) on a separate picture. The two pictures are equal, which gives $\sum F = ma$ and a moment equation about any point you like. Integrate that law along the path and you get energy; integrate it over time and you get momentum. Kinematics from Part I supplies whatever link between accelerations you are still missing.

Written as a study companion for MEC2047S Dynamics I, built on the course's lecture notes L01–L35, tutorials 0–5 and the two review lectures. Every worked number was re-computed, and each velocity and acceleration diagram is drawn to scale from those computed values. Where a slide or memo number did not reproduce, the text says so.